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Predicting Intensive Care Unit Length of Stay Using MIMIC-III Critical Care Database

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Abstract

Background: Stroke is a leading cause of global mortality and morbidity and a frequent reason for prolonged ICU stays. Prolonged ICU length of stay (LOS) poses various challenges such as increased risk of complications, strained healthcare resources, and elevated costs. Predicting LOS for stroke patients facilitates early rehabilitation, reduces risks such as hospital-acquired infections, and optimizes healthcare resource use. Machine learning (ML) offers the potential to process big data and discern intricate patterns, making it invaluable for healthcare applications. This study leverages the Medical Information Mart for Intensive Care III (MIMIC III) dataset to develop a reliable ML-based predictive model for ICU LOS in stroke patients. The model serves as a planning tool for resource allocation and as a benchmark for specialized care models, ultimately aiming to optimize hospital resources and elevate patient care quality.

Methods: The MIMIC-III dataset includes data of over 40,000 ICU patients from the Beth Israel Deaconess Medical Center between 2001 and 2012. Patients aged between 20 and above admitted to the ICU and diagnosed with diseases relevant to stroke were included. The study considered 49 predictors categorized under Demographic Information, Admission Details, Laboratory Tests, and Physical Examination. After data preprocessing, the dataset was split in an 80:20 ratio for 5-fold cross-validation and internal validation. The outcome measure, ICU LOS was dichotomized based on a 7-day threshold. Three classification algorithms, Logistic Regression (LR), Random Forest (RF), and Support Vector Machine (SVM), were used for the study. Evaluation metrics were accuracy and area under the curve (AUC).

Results: From the MIMIC-III dataset, 555 cases met eligibility criteria with a diagnosis of stroke and were included in the study. 72.07% of cases had ICU LOS less than 7 days. Univariable analysis showed a higher proportion of males and a majority being of White ethnicity were presented among the stroke population. RF model showed superior performance, achieving an accuracy of 0.82 with a standard error (SE) of 0.025 and an AUC of 0.82 with a SE of 0.037 during 5-fold cross-validation. Internal validation yielded an accuracy of 0.81 and AUC of 0.85.

Conclusions: ML models, especially the RF model, demonstrated promising potential in predicting ICU LOS for stroke patients using the MIMIC-III dataset. However, there are several limitations, including potential data imbalance, the exclusion of specific variables, and reliance on a single database. Addressing these issues in future research can further refine the model's predictive accuracy and applicability across diverse healthcare scenarios.

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Abbreviations

ACC	Accuracy
ADL	Activities of Daily Living
ALP	Alkaline Phosphatase
ALT	Alanine Aminotransferase
APACHE	Acute Physiology and Chronic Health Evaluation
AST	Aspartate Aminotransferase
AUC	Area Under the Curve
AUROC	Area Under the Receiver Operating Characteristic Curve
bpm	Beat per Minute
CART	Classification and Regression Tree
CBFL	Community-Based Federated Machine Learning
CDC	National Center for Chronic Disease Prevention and Health Promotion
cm	Centimetres
C^o	Celsius
DALYs	Disability-Adjusted Life Years
dL	Decilitres
DT	Decision Trees
ELM	Extreme Learning Machine
ER	Emergency Room
FiO_2	Fraction of Inspired Oxygen
FL	Federated Machine Learning
FPR	False Positive Rate

g	Gram
GB	Gradient Boosting
GCS	Glasgow Coma Scale
G-Mean	Geometric Mean
GRU	Gated Recurrent Unit
HDL	High-Density Lipoprotein
IADL	Instrumental Activities of Daily Living
ICD-9	International Classification of Diseases, Ninth Revision
ICU	Intensive Care Unit
INR	International Normalised Ratio
IQR	Interquartile Range
IU	International Unit
JITL	Just-In-Time Learning
kg	Kilogram
L	Liters
LDA	Linear Discriminant Analysis
LOS	Length of Stay
LR	Logistic Regression
LSTM	Long Short-Term Memory
mEq	Milliequivalents
mg	Milligram
MIMIC-III	Medical Information Mart for Intensive Care III
MIT	Massachusetts Institute of Technology
ml	Millimetre

mmHg	Millimetre of Mercury
mmole	Millimole
NN	Neural Networks
OASIS	Oxford Acute Severity of Illness Score
$PaCO_2$	Partial Pressure of Carbon Dioxide in Arterial Blood
PaO_2	Partial Pressure of Oxygen in Arterial Blood
PT	Prothrombin Time
PTT	Partial Thromboplastin Time
RBF	Radial Basis Function
RF	Random Forest
ROC	Receiver Operating Characteristic
SAPS	Simplified Acute Physiology Score
SD	Standard Deviation
SE	Standard Error
SQL	Structured query language
SVM	Support Vector Machine
SVR	Support Vector Regression
TIAs	Transient Ischemic Attacks
TPR	True Positive Rate
XGB	Extreme Gradient Boosting
μL	Microliters

Glossary

Term	Definition
Extract, Transform, and Load (ETL)	A systematic process used to gather data from various sources, refine it through transformation, and then store or "load" it into a database or data warehouse. This process is fundamental in data integration strategies, ensuring that information is accessible, consistent, and structured for analysis or reporting purposes.
Feature	A measurable property or characteristic of the object being analysed, such as an attribute of a data sample that can be used in making predictions. For example, blood glucose level is a feature commonly used to predict and manage diabetes in patients.
Internal Validation	A process of evaluating the model's performance using a subset of the data that the model has not seen during training.
Interquartile Range (IQR)	A statistical measure used to describe the spread of the middle 50% of a dataset. In this study, it is reported in the range of the first quartile (25th percentile) from the third quartile (75th percentile). The IQR provides insight into the dataset's variability, highlighting where most of the values lie.
Standard Deviation (SD)	A statistical measure that quantifies the amount of variation or dispersion in a set of values. A low SD indicates that the values tend to be close to the mean, while a high SD indicates that the values are spread out over a wider range. It's important to note that the interpretation of the SD depends on the scale of the measure, so it might not signify the same degree of dispersion in different contexts.
Standard Error (SE)	A statistical measure used to quantify the precision of an estimate, often the sample mean. A low SE indicates that the sample mean is a more accurate reflection of the population mean, while a high SE suggests less accuracy. Similar to SD, the interpretation of the SE is scale-dependent and may vary in significance across different contexts.

Introduction

Problem Statement

Efficient management of Intensive Care Unit (ICU) resources is essential for controlling the Length of Stay (LOS), a metric used in healthcare management. Especially in stroke cases, reducing ICU LOS allows for timely rehabilitation and therapy, reduces risks such as hospital-acquired infections, and optimizes healthcare resource use. Stroke, a leading global cause of death and disability, requires precise prediction of ICU LOS aids healthcare providers in optimising resource usage and improving patient outcomes.

Due to the complex variables influencing LOS in stroke cases, machine learning (ML) is an appropriate tool to process vast amounts of data and identify complex patterns, enhancing its potential in healthcare. This study focuses on the Medical Information Mart for Intensive Care III (MIMIC III) dataset, a publicly available database offering comprehensive ICU data to build a reliable predictive model for ICU LOS by targeting stroke patients.

Utilising ML techniques with the MIMIC III dataset allows for robust modelling, considering the wide-ranging variables in patient care. This study aims to contribute to the understanding and application of ICU LOS prediction, thereby offering healthcare providers data-driven insights to improve resource management and enhance the quality of care in critical medical conditions, such as stroke.

Background

Stroke and Burden of Disease

The definition of stroke was updated in 2013 by the American Heart Association and the American Stroke Association. It refers to a medical condition that affects central nervous system organs.[1] It occurs when blood flow is disrupted, either by a blockage in the blood vessels (an ischaemic stroke) or bleeding in the brain (a haemorrhagic stroke). Such disruptions deprive neuron cells of essential oxygen and nutrients, causing cell damage or death. The damage often inflicts lasting harm on the neuronal organs, potentially resulting in long-term disability or even death in stroke patients.[2] Even though the new definition of stroke includes the brain and other organs, such as the spinal cord and retinal cells,[1] this project only focuses on stroke with lesions on the brain.

Stroke ranks second among the causes of death, with approximately 5.5 million global fatalities annually.[3] According to the National Center for Chronic Disease Prevention and Health Promotion (CDC), the mortality rate for stroke in 2021 was approximately 41.1 per 100,000 in the United States.[4]

While the high mortality rate associated with stroke is a major concern, it often also leads to permanent disability, making it a leading cause of acquired disability. Dhamoon et al. (2017) conducted a study comparing disability events between stroke and myocardial infarction patients. Both groups exhibited similar vascular risk factors and were subject to sudden vascular events, such as vascular occlusions. Disabilities assessed through activities of daily living (ADL), which measure basic self-care tasks, and Instrumental ADL (IADL), gauging more complex life activities, revealed a more substantial increase score for stroke patients (0.88 points on a disability scale, where higher values indicate greater disability, with a 95% confidence interval of 0.57–1.20, $p<0.0001$) than myocardial infarction patients (0.20, 0.06–0.35, $p = 0.006$). [5] According to Lopez et al. (2001), stroke was the second leading cause of disease burden, as measured by Disability-Adjusted Life Years (DALYs), in high-income countries and the fifth leading cause in low-to-middle-income countries.[6]

Stroke can also lead to various complications, with post-stroke patients often needing support or continuous institutional care for their daily living. The post-stroke complications can vary greatly in type and severity, depending on the affected brain area. These complications may include medical issues, such as post-stroke seizures and urinary and bowel incontinence; musculoskeletal problems, such as muscle pain, spasticity, and hypertonicity; and psychological complications, such as post-stroke depression and changes in mood and emotion.[7]

Hankey et al. (2022) collected information from patients who had an acute cerebrovascular incident and managed to survive the initial 30 days between 1989 and 1990. The findings indicate that only half of these patients lived beyond five years, whereas a third suffered from permanent disability. Moreover, a seventh of these patients needed continuous institutional care for their daily living.[8]

Stroke and Intensive Care Unit Length of Stay

ICU LOS refers to the duration a patient spends in the hospital from ICU admission to discharge, with a same-day admission and discharge counting as a one-day stay. The duration of ICU LOS can vary widely, from less than a day to several months, depending on factors such as hospital type [9], patient condition [10–14], insurance claims [15], and hospital management. [10,12,15–21]

A very short ICU stay lasts less than 24 hours [12], whereas a prolonged stay could be considered an LOS that exceeds a specific number of days, depending on the study. Some studies use the 21- and 28-day benchmarks to indicate chronic critical illness. [13,14]

A systematic review by Langhorn et al. (2005) suggests that stroke patients admitted to the ICU experience a reduction in both mortality rate and poor outcomes compared to those in general wards.[22,23] Mortality and disability can increase with the complications experienced during the early phase of stroke, such as infection, seizures, and thromboembolism.[23] Therefore, for stroke patients, ICU admission may be necessary to facilitate specialised care and monitoring.

A patient's time in the ICU following a stroke is associated with critical clinical sequelae and high hospital costs and resource use. [24] Prolonged LOS increases costs and resource utilisation, making it a vital metric for evaluating management efficiency and hospitalisation cost. [25–27] An extended LOS is clinically associated with a higher risk of complications such as infection [28–30], falls [30,31], constipation [30,32], and mortality [33] in stroke patients. An expedited transition from the ICU to the general ward facilitates early rehabilitation and often improves outcomes. [34,35] Therefore, accurately predicting ICU LOS for stroke patients is essential for hospitals to optimise care and effectively manage resources.

Machine Learning

Introduction to Machine Learning

Machine Learning (ML) emerged from Arthur Samuel's 1959 research, ‘Some studies in machine learning using the game of checkers’. [36] Today, ML refers to a field of study that applies data and algorithms to make predictions or decisions. Unlike in a broader context in artificial intelligence, where the goal may be to replicate human learning processes, the focus of this project is using machine learning to predict ICU LOS in stroke patients. The algorithms used in this project, including logistic regression (LR), random forest (RF), and support vector machine (SVM), are part of the supervised learning category and are used specifically for a binary classification task. These predictive models are instrumental in achieving the project's objectives of leveraging data-driven insights to optimise decision-making.

Learning System

Machine learning is divided into three primary categories: supervised, unsupervised, and semi-supervised learning. Supervised learning requires labelled training data, which involves associating input data (such as demographic, vital signs, laboratory values, etc.) with corresponding target labels (such as health outcomes). The model attempts to determine the

relationship between the input and target labels, thus enabling predictions or classifications. [37] Common algorithms in this category, such as linear regression, LR, and RF, are used in this study. These techniques are often applied in regression and classification tasks, bridging the gap between data-driven insights and actionable medical or epidemiological conclusions.

Unsupervised learning involves learning from unlabelled data to discover patterns, structures, or relationships within it. This type includes clustering algorithms, such as k-means clustering, and dimensionality reduction techniques, such as principal component analysis. Applications include data exploration, anomaly detection, and recommendation systems.[38]

Semi-supervised learning combines the principles of both supervised and unsupervised learning and utilises both labelled and unlabelled data. It is beneficial when acquiring labelled data is costly or challenging but a substantial amount of unlabelled data exists.[39]

Supervised learning is the best approach for this project, as it allows for precise predictions and classifications using labelled data, aligning with the project's objectives and the available dataset.

Classification and Regression Task

Classification and regression are two fundamental types of models used in machine learning and statistical analysis.

Classification models predict categorical or discrete outcomes by assigning input variables to predefined categories or classes. Such models learn from labelled training data, where each data point is associated with a specific class. Various algorithms, such as decision trees (DT), SVM, or neural networks, perform classification. Notably, LR, despite the term ‘regression’, is considered a classification technique in the machine learning field. For example, whether to classify an email as spam based on content and features falls under this category.[37]

Regression models predict continuous numerical values by analysing the relationship between input variables and a continuous target variable. Unlike classification models, which predict categorical outcomes, regression aims to determine specific numerical outcomes. An example includes predicting house prices based on attributes like square footage, number of bedrooms, and location. Regression aims to minimise the difference between predicted and actual values by fitting a mathematical function to the data.[40]

Systematic Literature Review

The systematic literature review aimed to collate and summarise research into the prediction of ICU LOS in stroke patients that employs machine learning techniques. The goal was to assess the existing methods, data sources, and outcomes related to ICU LOS prediction in this specific patient population. The following criteria guided the publication selection:

1. Population: The study must only involve adult patients (aged 20 or above) admitted to the ICU following a stroke.

2. Study Design: The research must be

2.1 Predictive, employing machine learning, and using ICU LOS as an outcome.

2.2 Based on structured data for model training. Studies using imaging data or unstructured data, like doctors' or nurses' notes, were excluded.

2.3 Not use surgical data, such as perioperative and post-operative data, as training data.

3. Publication Type: Only published articles or poster presentations with original articles in English were considered.

Using the search keywords 'stroke', 'cerebrovascular disease', 'ICU', 'length of stay', and 'prediction' in the PubMed, Cochrane, Scopus, and Embase search engines until July 2023, a total of 141, 0, 62, and 97 search results were found, respectively. After eliminating 53 duplicate studies, 219 were selected for abstract screening. The following criteria guided the publication screening:

Following the above guidelines, 143, 75, and 1 studies were excluded due to not meeting eligibility criteria for the population, study design, and publication type criteria, respectively. Thus, only 2 studies remained. After a thorough review, only one study met all the requirements.

Therefore, for the second round of publication searches, the search criteria were broadened, particularly criterion 1, ‘Population’. Instead of focusing solely on adult patients admitted to the ICU following a stroke, this criterion was expanded to include studies whose populations were drawn from any patients admitted to the ICU, regardless of diagnosis. The other criteria remained the same.

From PubMed, Cochrane, Scopus, and Embase, 124, 0, 146, and 195 search results were obtained, respectively. After eliminating 175 duplicate studies, 290 were selected for abstract

screening. Following these guidelines, 16, 224, and 6 studies were excluded for not meeting eligibility criteria for the population, study design, and publication type criteria, respectively. Thus, 44 studies remained. After a thorough review of the full paper, 14 and 4 studies were excluded for not meeting the study design and publication type criteria, respectively, leaving only 26 articles for review. Table 1 presents the details of the studies included in this systematic review.

Author	Year	Prediction Model/Score	Best Model: Performance Metric	Predictors Variable	Data Source
Houthooft, R. et al.[41]	2015	SVM, SVR, ANN, RF, KNN, ADA	SVM: AUC 0.82 (Classification) SVR: MAE 1.79 (Regression)	SOFA scores, patient information, ICU stay information, ICU monitored values and lab test results from the first five days after admission	Ghent University Hospital (2009-2013)
Ling, Y. et al.[42]	2018	RF	AUC 0.86, sensitivity 0.88, specificity 0.71	Mechanical ventilation use, OASIS, LODS, GCS, SAPS II, SAPS III, vasopressor use, SOFA scores, urine volume, shock index	MIMIC-III
Balkan, B. et al.[43]	2018	RF	RF: MAE 2.010, R^2 0.016	APACHE IV and IVa input variables including numerical measurements, binary flags, and admission diagnosis	eICU Collaborative Research Database
Huang, L. et al.[44]	2019	CBFL, FL	AUC 0.65	Drugs administered on patients during the first 48 hours of ICU stay	eICU Collaborative Research Database
Ma, X. et al.[45]	2019	JITL-ELM, SVM	JITL-ELM: AUC 0.85, Precision 1.00, Accuracy 0.82, Specificity 1.00, Sensitivity 0.6150	Patient information, physiological parameters of each ICU patient in the first 48 hrs after their admission	MIMIC-III
Williams Batista, R. & Sanchez-Arias, R.[46]	2020	Tree-based methods (RF, Boosted Trees (XGB and C5.0)), LR	RF: Accuracy 0.6, Kappa 0.20	Demographic data, disease characteristics, prior health status, surgeries, care details, ICU types, transfer units, renal function, diagnostic ICD-9 codes	MIMIC-III
Wu, J.Y., et al.[47]	2021	SVM, CART, RF	RF: AUC 0.77	Age, heart rate, systolic blood pressure, SAPS-II, temperature, PaO_2/FiO_2	MIMIC-III
Su, L., et al.[48]	2021	LR, RF, XGB	RF: AUC 0.76	Key medical data from the first 6 hrs in the ICU in sepsis patients	ICU of Peking Union Medical College

Author	Year	Prediction Model/Score	Best Model: Performance Metric	Predictors Variable	Data Source
					Hospital, China (2016-2018)
Wu, J., et al.[49]	2021	RF, SVM, Deep learning, GB	GB: AUC 0.74	Age, chronic diseases, type of admission, measurements during the first 24 hrs in ICU (including vital signs, Pao2/Fio2 ratio, urinary output, lab test, GCS)	eICU Collaborative Research, MIMIC-III
Abd-Elrazek, M.A., et al.[50]	2021	NN, Classification Trees, Tree Bagger, RF, Fuzzy Logic, SVM, KNN, Regression Tree, Naive Bayes	Fuzzy Logic: Accuracy 0.92	Demographic data, vital signs, laboratory tests, GCS, chronic illness status, current complaint, diagnosis	Suez Canal University Specialized Hospital ICU, Egypt (2015-2017)
Alghatani, K., et al.[51]	2021	Linear Regression, LDA, RF, KNN, SVM, XGB, Multiple Linear Regression, SVR	RF: Accuracy 0.65	Demographic data, vital signs, extra features engineered from baseline vital signs (modified means, standard deviations, quantile percentages)	MIMIC-III
Kurtz, P., et al.[52]	2022	SVR, KNN, GB, RF, CART, bagging, linear regression	RF: AUC 0.73 (Classification) RF: RMSE 8.3, MAE 6.54 (Regression)	Demographics data, ICU admission diagnosis, comorbidities, acute illness severity scores (SOFA, SAPS 3, GCS, etc.), use of organ support in stroke patients	Data from 130 ICUs from 43 Brazilian hospitals (2011-2020)
Alsinglawi, B., et al.[53]	2022	RF, XGB, LR	RF: AUC 0.98	Demographic data, diagnosis, laboratory features in lung cancer patients	MIMIC-III
Hu, Y., et al.[54]	2022	NN, GB	NN: R^2 0.40 (Regression) GB: Accuracy 0.82 (Classification)	Demographic data, admission information, diagnoses, lab tests, ICU charted observations within first 8 hr of admission in diabetic patient	MIMIC-III
Faria, M., et al.[55]	2022	DT, RF, KNN, GB	RF: AUC 0.9997	Demographic data and clinical records	Centro Hospitalar do Tâmega e Sousa, Portugal (2017-2021)
Iwase, S., et al.[56]	2022	RF, XGB, NN, LR	RF: AUC 0.88 (≤ 1 week)	Demographic data, diagnosis, admission route,	Chiba University

Author	Year	Prediction Model/Score	Best Model: Performance Metric	Predictors Variable	Data Source
			RF: AUC 0.89 (> 2 week)	comorbidity within 24 hr after ICU admission	Hospital, Japan (2010-2019)
Al-Dailami A., et al.[57]	2022	TCNs	AUC: 0.91	Demographic data, diagnosis, time-series features laboratory test, vital signs	eICU Collaborative Research, MIMIC-IV
Peres, I.T., et al.[58]	2022	SVR, KNN, GB, RF, Bagging, Linear Regression, Generalized Linear Model, CART	Ensemble of RF with LR: RMSE 3.82, MAE 2.52, R^2 0.36	Demographic data, Comorbidities, Complications, Laboratory data, Primary and secondary diagnosis	109 mixed-type ICUs and 93 medical-surgical ICUs from Brazilian hospitals
Bednarski, B.P., et al.[59]	2022	TCNs, LR, RF, GRU	GRU-D: AUC 0.72 ± 0.02 (< 3 days) RF: AUC 0.75 ± 0.04 (> 7 days)	Time-series vital signs data, 312 clinical objective features for the first 24 hrs of admission	MIMIC-III
Deng, Y., et al.[60]	2022	Recurrent Neural Network (RNN), GRU, LSTM	LSTM: AUC 0.77	Variables collected during the initial 24 hrs after ICU admission based on APACHE II, SOFA, and SAPS II scoring systems	MIMIC-IV
Alabbad, D.A., et al.[61]	2022	RF, GB, XGB, Ensemble models	RF: Accuracy 0.94	Demographics data, comorbidities, laboratory results, symptoms, vital signs in COVID-19 patient	King Fahad University hospital, Saudi Arabia
Al-Mamun, M.A., et al.[62]	2022	LR	AUC 0.82	Variables in Medication Regimen Complexity Index	ICU in community hospital, USA (2022)
Boussen, S., et al.[63]	2022	Gaussian mixture model	Accuracy 0.8, AUC 0.87	Breathing frequency, SpO2 in COVIC-19 patient	Aix-Marseille Université Faculté de Médecine, France (2020-2021)
Alsinglawi B., et al.[64]	2022	Ensemble model, Deep learning, GBR, Staking Regressor, RF	GB: R^2 0.81±0.07	Demographics data, ICU admissions, vital signs, laboratory tests, medications of heart failure patient.	MIMIC-III
Sotoodeh M., et al.[65]	2022	Hidden Markov model and 6 baseline regression model	Linear regression: RMSE: 225.05	Vital signs, GCS, serum glucose, white blood cell count, urine output, variables from SAPS II score for linear regression	Physionet 2012 challenge dataset

Author	Year	Prediction Model/Score	Best Model: Performance Metric	Predictors Variable	Data Source
González-Nóvoa J, et al.[66]	2023	XGB	MAE 2.529	Demographic Information, GCS, Vital Signs, Laboratory Results	MIMIC-III

Table 1 Information from the publication predicting ICU LOS according to the systematic review. Text in bold indicates the study that exclusively targeted stroke patients (Kurtz et al., 2022).

APACHE = Acute Physiology and Chronic Health Evaluation, AUC = Area Under the Curve, CART = Classification And Regression Tree, CBFL = Community-Based Federated Machine Learning, DT = Decision Tree, ELM = Extreme Learning Machine, FL = Federated Machine Learning, FiO_2 = Fraction of Inspired Oxygen, GB = Gradient Boosting, GCS = Glasgow Coma Scale, GRU = Gated Recurrent Unit, JITL = Just-In-Time Learning, KNN = K-Nearest Neighbour, OASIS = Oxford Acute Severity of Illness Score, PaO2 = Partial pressure of oxygen in arterial blood, LDA = Linear Discriminant Analysis, LODS = Logistic Organ Dysfunction, LR = Logistic Regression, LSTM = Long Short-Term Memory, NN = Neural Networks, RF = Random Forest, RMSE = Root Mean Square Error, SAPS = Simplified Acute Physiology Score, SOFA = Sequential Organ Failure Assessment, SVM = Support Vector Machine, SVR = Support Vector Regression, TCNs = Temporal Convolutional Networks, XGB = Extreme Gradient Boosting

As given in Table 1, various studies have proposed different models for ICU LOS prediction. MIMIC-III was one of the most utilised training datasets, followed by eICU Collaborative Research Database and university hospital dataset, which mostly target a general patient population instead of focusing specifically on stroke patients. Some studies focus on diseases such as sepsis[48], lung cancer[53], and diabetes [54], and a more recent work has focused on COVID-19 patients. [61,63]

The work of Kurtz et al., 2022[52] is notable, as they exclusively targeted stroke patients, similar to the current study. The work comprised both ICU LOS regression (prediction of the actual number of days for ICU LOS) and classification tasks (less than or equal to 14 days, or more than 14 days) while utilising models including support vector regression (SVR), K-Nearest Neighbours (KNN), gradient boosting (GB), RF, and bagging. RF outperformed the rest in both regression and classification with area under the curve (AUC) = 0.73 for classification, RMSE = 8.3, and MAE = 6.54 for regression task. The features included demographic information, ICU admission diagnosis, comorbidities, and medical assessment scores. Nevertheless, the study did not utilize vital signs or laboratory test data as training sets.

However, in the other works, the models utilise a wide range of variables, from demographic data and clinical scores to lab results, underlining the multifaceted nature of ICU LOS prediction. Notably, the Glasgow Coma Scale (GCS), a common scoring system used to assess a patient's consciousness[67], holds particular significance for stroke patients, and offers insights into stroke severity and potential outcomes.[68,69]

Numerous models, including deep learning, have been used to explore ICU LOS prediction. Classification tasks are commonly preferred over regression tasks for LOS prediction, with many studies suggesting the RF model as having the best performance. AUC frequently serves as the primary performance metric for classification, whereas mean absolute error (MAE) is often employed for regression tasks. However, the optimal model performance depends on the dataset, with AUC values varying from 0.65[44] to 1. [55]

Despite extensive research on ICU LOS prediction, a noticeable gap exists in studies dedicated to stroke patients. The only related study, which was performed by Kurtz, P., et al. (2022), was conducted in Brazil. This geographic focus raises concerns about the generalisability of their findings to other regions or healthcare settings. Moreover, with an AUC of 0.73 for classification and reported metrics of RMSE 8.3 and MAE 6.54 for regression, the study suggests the potential for enhancing prediction accuracy when compared to other research.

Kurtz, P., et al. (2022) incorporated features such as demographic data (age, gender), ICU admission diagnosis (ischemic or haemorrhagic stroke), comorbidities, and acute illness severity scores. However, the MIMIC III database provides a more detailed diagnosis using ICD-9 codes, initial admission laboratory results, and vital signs, as well as additional demographic information, such as ethnicity and insurance status. Utilising these comprehensive predictors could improve the prediction model's accuracy and applicability.

In summary, while the existing study provides valuable insights, there are several areas where further research, especially using comprehensive datasets like MIMIC III, may offer improvements in ICU LOS prediction for stroke patients.

Aim

This project aims to explore the predictive power of machine learning models in predicting the LOS in the ICU among stroke patients using inpatient data from the MIMIC-III Critical Care Database and compare their performance to find the best LOS prediction.

Objective

1. Extract data from the MIMIC-III Critical Care Database using structured query language (SQL) programming, ensuring a reproducible workflow.
2. Examine associations between predictors and explore the relationships between each predictor and the ICU length of stay using data visualisation and analysis tools in Python and R.
3. Develop predictive models for ICU length of stay using various supervised learning algorithms and compare their performance using suitable metrics.
4. Evaluate the generalisability and applicability of the developed models on separate subsets of the MIMIC database.

Material & Methods

MIMIC-III Dataset

The Medical Information Mart for Intensive Care III (MIMIC-III) project is a large, freely available database of de-identified health-related data associated with over 40,000 patients who stayed in critical care units of the Beth Israel Deaconess Medical Center between 2001 and 2012. The project was developed by the Laboratory for Computational Physiology at the Massachusetts Institute of Technology (MIT) in collaboration with the Beth Israel Deaconess Medical Center.[70]

The MIMIC-III project aims to provide researchers with a large dataset to study the clinical care of critically ill patients. The data can be used to support a wide range of research studies, including epidemiological studies, clinical decision-making, and predictive modelling. The MIMIC-III database contains a variety of data types, including demographic information, vital sign measurements, laboratory test results, medications, and clinical notes. The data were collected from a variety of sources, including electronic health records, hospital information systems, and bedside monitors, and have been de-identified to protect patient privacy.

The potential of the MIMIC-III dataset for LOS prediction has been demonstrated in various studies. However, no studies have utilised the MIMIC-III dataset to evaluate LOS prediction for stroke patients, and this warrants further investigation.

Extracting Data

This study focuses on a specific population: individuals aged 20 years and older who were admitted to the ICU at least once during their hospital stay due to an urgent or emergency admission. The primary diagnosis for these individuals must fall within the range of specific International Classification of Diseases, Ninth Revision (ICD-9) codes related to group of cerebrovascular diseases, including stroke.

Inclusion Criteria

1. The patient's age at the time of hospital admission must be between 20 and above.

2. The patient must have been admitted to the ICU at least once during their hospital stay on that visit, and the type of admission must be either 'emergency' or 'urgent'.

3. The patient must have the following ICD-9 code in their records for the primary diagnosis. The code should fall between 430 and 437, which corresponds to the cerebrovascular disease group. However, the study excludes code 435, which is related to transient ischemic attacks (TIAs), which are reversible. Refer to Table 2 for ICD code details.

ICD-9 Code	Description
430	Subarachnoid haemorrhage
431	Intracerebral haemorrhage
432	Other and unspecified intracerebral haemorrhage
433	Occlusion and stenosis of precerebral arteries
434	Occlusion of cerebral arteries
436	Acute but ill-defined cerebrovascular disease
437	Other and ill-defined cerebrovascular disease

Table 2 ICD-9 codes and descriptions relevant to the primary diagnosis of stroke for this project.

Exclusion Criteria

1. This study focuses on the cerebrovascular disease group, specifically permanent brain vascular pathology. Therefore, cases involving TIAs, precerebral, or vertebral vascular pathology were removed.

2. Records of patients who died during hospital admissions were excluded.

Variable Description

The extracted features fall into four categories: Demographic Information, Admission Details, Laboratory Test (which includes Complete Blood Count, Coagulation, Clinical Chemistry, Lipid Profile, Arterial Blood Gas), and Physical Examination, totalling 49 variables. Refer to Table 3 for details. However, not all these variables are final, as the next step involves removing any that exceed the missing value limit. Only the initial data from Laboratory Tests and Physical Examinations within 24 hours of ICU admission were collected for this study to ensure consistency and represent the patient's initial condition, minimising the influence of extended hospitalisation effects.

Category	Variable Name	Class	Label	Description	Unit
Demographic Information	Age Group	Categorial	20-29	Age was calculated by subtracting the birth date from the admission date and then grouped into seven categories. Due to the MIMIC-III study's data privacy protocols, birth dates for patients over 80 were modified to prevent identification, in	Age: Year
			30-39		
			40-49		
			50-59		
			60-69		

Category	Variable Name	Class	Label	Description	Unit
			70-79	compliance with HIPAA regulations. Consequently, these patients' ages are recorded as over 300 years in the database. Thus, subjects over 80 years at hospital admission are represented as over 300 years, obscuring their exact age.[70]	
			over 80		
	Gender	Categorial	Male		
			Female		
	Ethnicity Grouped	Categorial	Asian	39 different ethnicities were grouped into six main categories.	
			Black		
			Hispanic		
			Other		
			Unknown		
			White		
	Insurance	Categorial	Government	Type of insurance	
			Medicaid		
			Medicare		
			Private		
			Self-Pay		
Admission Details	Admission type	Categorial	Emergency		
			Urgency		
	Admission diagnosis	Categorial	430	Only the primary diagnosis for admission was included in this column. Diseases were categorized based on the main ICD-9 groups.	
			431		
			432		
			433		
			434		
			436		
			437		
	History of Previous Stroke	Categorical	0 (No)	Identified from previous hospital admissions with a registered ICD-9 code starting with 43.	
			1 (Yes)		
	ICU LOS	Categorical	≤ 24 hours	The variable served as the model prediction result. Further detail is described in the 'Labelling Result' section.	LOS: Day
			Within 1 week		
			Within 2 weeks		
			≥ 14 days		
	Time from ER to ICU	Numeric		The variable calculated by subtracting the ICU admission time from the admission datetime.	Minute
	Respiratory Support	Categorical	0 (No)	The variable indicated the use of respiratory support during ICU admission. If a procedure identified by an ICD-9 code beginning with 96 is recorded during admission. It was	
			1 (Yes)		

Category	Variable Name	Class	Label	Description	Unit
				labelled '1' for 'Yes', and '0' for 'No' if not present.	
Laboratory Test: Complete Blood Count	Haematocrit	Numeric		Refers to a proportion of red blood cells in whole blood.	%
	Haemoglobin	Numeric		Refers to an amount of haemoglobin, a protein in red blood cells, in whole blood.	g/dL
	White Blood Cells	Numeric		Refers to an amount of white blood cells in whole blood.	x 10^3/μL
	Neutrophils	Numeric		Refers to a proportion of neutrophils cells in white blood cells.	%
	Lymphocytes	Numeric		Refers to a proportion of neutrophils cells in white blood cells.	%
	Platelet Count	Numeric		Refers to an amount of platelet in whole blood.	x 10^3/μL
Laboratory Test: Coagulation	INR(PT)	Numeric		Ratio used to determine the clotting tendency of blood.	
	PT	Numeric		Value used to determine the clotting tendency of blood.	Second
	PTT	Numeric		Value used to determine the clotting tendency of blood.	Second
Laboratory Test: Clinical Chemistry	Albumin	Numeric		Level of a protein that maintains oncotic pressure, essential for balancing body fluid distribution between blood vessels and tissues	g/dL
	ALT	Numeric		Level of an enzyme found primarily in the liver and kidney, increased with liver damage.	IU/L
	ALP	Numeric		Level of an enzyme found primarily in the liver and bone, increased with liver damage or bone disorder.	IU/L
	AST	Numeric		Level of an enzyme found primarily in the liver and muscle, increased with liver damage.	IU/L
	Bicarbonate	Numeric		Level of bicarbonate, which assess and monitor acid-base and electrolyte status in blood.	mEq/L
	Bilirubin: Total	Numeric		Level of bilirubin in blood, used for checking liver condition	mg/dL
	Chloride	Numeric		Level of chloride, which assess and monitor acid-base and electrolyte status in blood.	mEq/L
	Creatinine	Numeric		Level of creatinine in blood, use to measure kidney function.	mg/dL
	Glucose	Numeric		Level of glucose in blood.	mg/dL

Category	Variable Name	Class	Label	Description	Unit
	Haemoglobin A1c	Numeric		Level of blood sugar attached to hemoglobin, used to measures average blood sugar levels over the past 3 months	%
	Lactate	Numeric		Level of lactate in blood, which is typically caused by impaired tissue oxygenation	mmol/L
	Potassium	Numeric		Level of potassium, which assess and monitor acid-base and electrolyte status in blood.	mEq/L
	Sodium	Numeric		Level of sodium, which assess and monitor acid-base and electrolyte status in blood.	mEq/L
	Urea Nitrogen	Numeric		Level of urea nitrogen in blood, which is a waste product that kidneys remove from blood.	mg/dL
Laboratory Test: Lipid Profile	Cholesterol: HDL	Numeric		Level of HDL cholesterol in blood.	mg/dL
	Cholesterol: LDL	Numeric		Level of LDL cholesterol in blood.	mg/dL
	Cholesterol: Total	Numeric		Level of total cholesterol in blood.	mg/dL
	Triglyceride	Numeric		Level of total triglyceride in blood.	mg/dL
Laboratory Test: Arterial Blood Gas	$PaCO_2$	Numeric		Level of carbon dioxide in the arterial blood.	mmHg
	PaO_2	Numeric		Level of oxygen in the arterial blood.	mmHg
	pH	Numeric		Level of pH in the arterial blood.	
Physical Examination at Admission	Admission Weight	Numeric		Weight measured at the time of admission.	kg
	Admission Height	Numeric		Height measured at the time of admission.	cm
	GCS: Eye Opening	Ordinal	4	Eyes opening spontaneously	
			3	Eye-opening to speech	
			2	Eye-opening in response to pain	
			1	No eye-opening	
	GCS: Verbal Response	Ordinal	5	Oriented conversation	
			4	Confused conversation	
			3	Inappropriate words	
			2	Incomprehensible sounds	
			1	No verbal response	
	GCS: Motor Response	Ordinal	6	Obeys commands	
			5	Localizes painful stimuli	
			4	Withdraws from painful stimuli	
			3	Abnormal flexion to painful stimuli (decorticate response)	
			2	Extension to painful stimuli (decerebrate response)	

Category	Variable Name	Class	Label	Description	Unit
			1	No motor response	
	Heart Rate	Numeric		Number of times the heart beats per minute.	bpm
	Diastolic Blood Pressure	Numeric		The lowest pressure in the arteries when the heart is at rest, measured using non-invasive equipment	mmHg
	Systolic Blood Pressure	Numeric		Peak pressure in the arteries when the heart is contracting, recorded with a non-invasive device.	mmHg
	Oxygen Saturation	Numeric		Ratio of oxygen-bound haemoglobin to the total amount of haemoglobin, measured using non-invasive equipment	%
	Temperature			Body temperature	C°

Table 3 Extracted variables from MIMIC-III and their information.

ALP = Alkaline Phosphatase, ALT = Alanine Aminotransferase, AST = Aspartate Aminotransferase, bpm = beat per minute, C° = Celsius ,cm = centimetres, dL = Decilitres, ER = Emergency room, g= Gram, GCS = Glasgow Coma Scale , HDL = High-density lipoprotein, INR = International normalized ratio, IU = International unit, kg = Kilogram, L = Liters, LOS = Length of stay, LDL = Low-density lipoprotein, mEq = milliequivalents, mg = Milligram, ml = Millimetre, mmHg = Millimetre of mercury, mmole = Millimole, $PaCO_2$ = Partial pressure of carbon dioxide in arterial blood, PaO_2 = Partial pressure of oxygen in arterial blood, PT = Prothrombin time, PTT = Partial thromboplastin time, µL = Microliters.

Handling Missing Values

Variables listed in Table 3 with more than 20% missing values were excluded from the analysis. Separately, ICU admission records were also excluded if they had more than 20% missing values for the listed variables.[71] In addition to removing cases and variables, single imputation method was applied. For normally distributed numerical variables, missing values were replaced with the mean. For non-normally distributed numerical variables, the median was used to fill in missing values. And for categorical variables, the mode replaced missing values. The normality of the distribution for the numerical variable was tested by Shapiro–Wilk test.

Shapiro–Wilk test

The Shapiro–Wilk test is a statistical technique used to assess whether a given set of numeric data is normally distributed.[72] It posits a null hypothesis that the sample is drawn from a normal population. If the p-value obtained from the test is less than the significance level, typically 0.05, the null hypothesis is rejected, suggesting the data might not be normally distributed. For this study, a p-value of <0.05 was used to determine a non-normally distributed outcome.

Processing Data

Due to the numerical input requirement for all the models (LR, RF, and SVM), the categorial variables were transformed into a numerical format with the one-hot encoding technique.

One-hot encoding

One-hot encoding, also known as dummy coding, is a process used to represent categorical variables as binary vectors in machine learning models.[73] This technique converts categorical variables into a numerical format for algorithmic analysis and prediction. For example, if a variable has three categories (A, B, and C), it would be encoded into three binary features: A, B, and C. Each feature has a value of '1' if the original variable belongs to that category and '0' otherwise, allowing the categorical variable to be represented as a sparse binary vector for each observation.[73]

Outcome Labelling

The result from the prediction model was ICU LOS measured in days and calculated by subtracting the ICU admission time from the ICU discharge time. If both times are identical, the ICU LOS is designated as 'less than 24 hours'. Our literature review revealed that no standard categorization for ICU LOS currently exists, as the cut point of prolonged ICU LOS could be ranged between 3 to 14 days.[52,59] Therefore, the wide-ranging ICU LOS was stratified into four categorical labels for descriptive analysis:

- Less than 24 hours
- Within 1 week (1 ≤ ICU LOS ≤ 6 days)
- Within 2 weeks (7 ≤ ICU LOS ≤ 13 days)
- More than 14 days

A 7-day period served as the prolonged ICU LOS cut point for model classification, dividing data into binary responses: 'less than 7 days' and ‘7 days or more', based on the ICU LOS distribution in this study and the literature review. Further details on the descriptive analysis for ICU LOS can be found in the Results section.

Dataset Splitting for Model Validation

After data processing, the final dataset was divided at a ratio of 80:20 for 5-fold cross-validation (training dataset) and an internal validation (testing dataset). Internal validation is a separate testing dataset used to evaluate the model's performance on unseen data. During data splitting, the ratio between cases of ICU LOS less than 7 days and those predicting 7 days or more was held at equally, both in the overall split and within each cross-validation fold.

Maintaining a consistent ratio when splitting data for classification prediction is crucial, especially when dealing with imbalanced datasets. If the ratio isn't preserved, the model might be overrepresented by the dominant class, making it biased and reducing its predictive power for the minority class, which can falsely inflate model accuracy since it might primarily predict the majority class. By ensuring consistent ratios, both training and validation sets mirror the original data's distribution, enabling the model to be trained and evaluated on balanced samples from each class. This method enhances the model's reliability and generalizability.

Descriptive Analysis

Univariable Analysis

Once the data had been processed, univariable analysis was performed for all variables. Numerical data that approximately following a normal distribution were summarised using mean and standard deviation (SD). Conversely, numerical data that did not follow a normal distribution were presented with a median and an interquartile range (IQR) from P_{25} to P_{75}. Categorical variables were tabulated.

Correlation Test

This process explains the relationships between the features and ICU LOS (both categorical and numerical), as well as among the features themselves. To evaluate this, Spearman's correlation was calculated pairwise between features and between outcomes and features for numeric and ordinal variables, with a cutoff point of 0.8 taken to indicate a strong association.[74] Additionally, the Cramer V statistic was employed for categorical variables. Although it lacks a specific cutoff, a value approaching 1 typically signals a very strong association.[75] All measures of correlation were represented in a heatmap.

Classification Algorithms

The classification task in the present study utilised three selected classification algorithms: LR, RF, and SVM. LR served as a foundational model, often used as a baseline in prediction tasks, facilitating a fundamental performance comparison. RF was selected for its demonstrated success in predicting ICU LOS in supervised learning contexts, particularly for stroke patients.[52] Given its capacity to manage large datasets with numerous variables, it was deemed suitable for this task. SVM, a staple in many supervised machine learning studies, was also chosen. The literature review revealed that SVM excels in early ICU LOS prediction work[41], underscoring its continued pertinence and effectiveness in the domain.

Logistic Regression (LR)

LR is a statistical modelling technique used for binary classification tasks to predict the probability of an event, in this case, the ICU LOS is equal or more than 7 days or not. LR assumes that the relationship between the input variables and the log odds of the event follows a linear relationship.

Log Odds Function

Odds is the ratio of the probability of the event of interest (in this case, ICU LOS equal or more than 7 days) to its complementary probability. The log odds function, also known as the logit function, refers to the logarithm of these odds.[76] Maximum likelihood estimation is used to estimate the coefficients that best fit the data and minimise the difference between the predicted probabilities and the actual outcomes.

The equation is defined as

$$log\frac{p}{(1-p)} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_m x_m$$

where p denotes the probability of ICU LOS exceeding 7 days. β_m is the regression coefficient, indicating the change in the log odds for a one-unit change in the corresponding the predictor variable. x_m represents the explanatory variables whereas m stands for the total number of explanatory variables in the model, and it can take values from 1 to m.[77] β_0 captures the baseline log odds when all explanatory variables are set to their reference level.[77]

In LR, a threshold converts predicted probabilities p into binary class labels. In this study, the threshold of the model was chosen after comparing the overall performance between the optimal threshold and the default threshold, which set at 0.5. Probabilities above this value predict ICU LOS equal or more than 7 days, whereas those below predict the opposite.

Finding the Optimal Threshold Using Geometric Mean

While a threshold of 0.5 is common, it might not perform optimally in many medical contexts where the balance between sensitivity and specificity is crucial. One alternative is to use the geometric mean (G-mean), which is the square root of the product of sensitivity (true positive rate) and specificity (true negative rate).

$$G - mean = \sqrt{Sensitivity\ x\ Specificity}$$

For various threshold values t (ranging from 0 to 1), calculate the sensitivity and specificity. This is usually done by classifying all instances with a predicted probability greater

than or equal to t as the positive class, in this case, having 7 days of ICU LOS or more, and the rest as the negative class, in this case, having less than 7 days of ICU LOS. Calculating the sensitivity and specificity based on the classifications. For each threshold t, calculate the G-mean using the formula mentioned above. The optimal threshold is the one that maximizes the G-mean.

Log Loss

Log loss, also known as logistic loss or cross-entropy loss, quantifies the discrepancy between the predicted probabilities and the observed outcome (ICU LOS equal or more than 7 days) of the data. The log loss function is defined as the negative logarithm of the predicted probability for the true class.[78] The log loss equation can be derived from likelihood used in MLE by taking the log and multiplying it by -1, which has the following final form:

$$Log\ loss = -\log(L(\beta))$$

$$= \sum_{i=0}^{I} (y_i\, ln(p_i) + (1 - y_i)\, ln(1 - p_i))$$

where $L(\beta)$ represents the likelihood function and β refers to the vector of parameters or coefficients of the model, p_i refers to the predicted probability of the i observation belonging to the positive class and y_i refers to an actual outcome of observation i.

Maximum Likelihood Estimation (MLE)

MLE is a statistical method used in LR to estimate the model's parameters, specifically the coefficients $(\beta_0, \beta_1, \ldots, \beta_m)$, by maximising the likelihood of observing the given data.[79] This approach, widely used for fitting LR models, involves iteratively adjusting the values of the coefficients using optimisation algorithms such as gradient descent or Newton–Raphson to find the optimal values.[79] Once the MLE estimates are obtained, they can be used to calculate the log odds of the outcome based on the input variables and the application of the logistic function to obtain the predicted probabilities. The likelihood function equation can be written as

$$L(\beta) = \prod_{k:y_i=1} p_i \prod_{k:y_i=0} (1 - p_i)$$

where $L(\beta)$ represents the likelihood function and β refers to the vector of parameters or coefficients $(\beta_0, \beta_1, \ldots, \beta_m)$ of the model, p_i refers to the predicted probability of the ith patient belonging to the positive event category (ICU LOS equal or more than 7 days) and y_i refers to an

actual outcome of the ith patient. It takes the value of 1 if the patient experiences the event (ICU LOS equal or more than 7 days) and 0 otherwise.

Classification Trees

A tree-based classifier is structured with nodes and branches. A node represents a feature from the training data and serves two primary purposes: splitting the data based on certain criteria or providing a result when a specific path in the prediction process is reached. The initial point of a decision tree is called the root node. Branches, on the other hand, are paths leading from one node to another, representing decisions based on the features. The terminal nodes, or leaf nodes, signify the final prediction or classification.

Classification Criteria

Tree-based classifier algorithms require classification criteria to divide data into subgroups. Such criteria determine the choice of feature for a node, the split of the node, and the measurement of accuracy, among other factors. Three primary criteria are utilised in this context: entropy, Gini index, and information gain.

Decision Tree (DT)

A DT simplifies complex decision-making by segmenting it into more manageable steps. In DT, the decision-making process is visualised in a tree-like structure. Each node in the tree represents a test on an attribute, which then branches out based on the result of that test. For instance, in a DT for predicting weather suitability for a picnic, nodes could test conditions such as ‘is it raining?’ or ‘is the temperature above 20°C?’, leading to outcomes such as ‘not suitable for picnic’ or ‘suitable for picnic’. Refer to Fig 1 for an illustration.

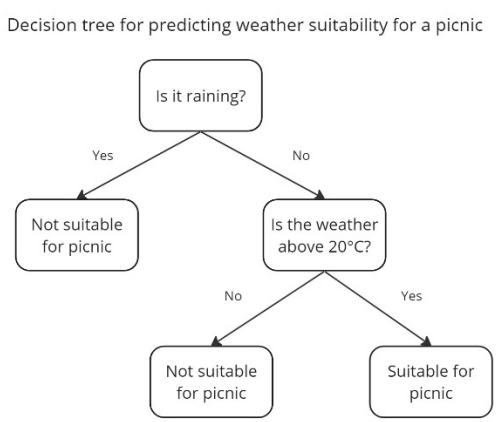

Figure 1 Decision tree diagram for predicting picnic weather suitability.

While DT offers advantages such as ease of comprehension and interpretability, it also has shortcomings. For instance, it has the propensity to overfit by creating complex, highly specific rules that may not generalise well to unseen data, resulting in poor predictive performance on new data.[80] Another disadvantage is the lack of smoothness in prediction, as DT produces piecewise-constant prediction rules, causing predicted values to change abruptly at decision boundaries.[81]

RFs, on the other hand, address some of these disadvantages by combining multiple DTs and reducing overfitting through ensemble learning.

Gini Index

Gini index, or Gini impurity, measures the likelihood of an incorrect classification of a random selection – in this case, a patient from the dataset. It is calculated by subtracting the sum of the squared probabilities of each outcome, which, in this case, is whether ICU LOS is equal or more than 7 days or not, from the total dataset. For binary classification, the Gini index value ranges between 0 and 0.5, where 0 expresses pure classification, wherein all elements belong to one outcome. On the other hand, the value of 0.5 suggests an equal distribution of elements over two outcomes. In DT modelling, the root node is selected based on the feature with the lowest Gini Index value. Subsequent nodes are determined by optimising the Gini index for the remaining data, with each node being chosen based on the next lowest Gini index value.

The Gini Index could be calculated as follows

$$Gini\ Index\ =\ 1 - \sum_{i\,=\,1}^{n} (P_i)^2$$

where n represents the number of interested outcomes or labels and P_i is the probability of an element being classified for each outcome.

Ensemble Learning

Ensemble learning is a method that combines multiple classifiers, often referred to as weak classifiers, to enhance machine learning algorithm performance. These classifiers are models that, on their own, perform marginally better than random guessing but can achieve greater accuracy when combined. While models such as regression and decision trees can sometimes act as weak classifiers, they can also be strong classifiers depending on the context. Another one example of weak classifier is decision stumps, which is a one-level DT, where the

root node directly connects to its terminal nodes, which represent the final decision or outcome. Ensemble methods include techniques like bagging, boosting, stacking, and others.

Bagging

Bagging, short for bootstrap aggregating method is used in RF. It involves creating multiple versions of a model by training on different bootstrap samples of the original training set, resulting in a diverse set of models.[82] A bootstrap replicate is a random sample drawn with replacement from the original training set.

The aggregation step in bagging depends on the prediction type. For numerical outcomes, the predictions from different models are averaged, reducing variance and improving accuracy.[82] For classification tasks, a plurality vote is applied, where the final prediction is given by the class predicted by the majority of models, and the class predicted by the majority of models is chosen as the final prediction.[82]

Random Forest (RF)

RF was first proposed by Ho (1995).[83] It is an ensemble learning method that aggregates results from several DTs. Instead of relying on a single tree's output, RF combines outputs from multiple trees, enhancing the robustness of predictions. Specifically, for classification tasks, RF outputs the mode from each DT, and for regression tasks, it provides the mean prediction of the outcome from each DT. The term 'forest' reflects the use of multiple DTs in the method. In this study, the Gini index was chosen as the criterion for the RF model due to its effectiveness in large data distributions and its superior straightforward implementation.

Support Vector Machine (SVM)

SVM is a set of supervised learning methods used for classification, regression, and outlier detection.[84] SVMs are based on the concept of decision planes, which define decision boundaries. In SVM, the principle arises from the idea that an appropriate plane can linearly separate data into two groups. In this study, linear SVM is used for developing a binary classifier.

Hyperplane

A hyperplane refers to a subspace with a dimension one unit less than its surrounding space. For instance, if considering a 3-dimensional space, the hyperplanes correspond to 2-dimensional planes. In a 2-dimensional space, the hyperplanes equate to 1-dimensional lines. Therefore, in an d-dimensional plane, the hyperplanes have dimensions of d-1.

The equation of a hyperplane, given a set of points x (where x is a vector in d-dimensional space), can be expressed as

$$\vec{w} \cdot x + b = 0$$

In this formula, $\vec{w}$ symbolises a normal vector perpendicular to the hyperplane.

Margin

Assuming the training data are linearly separable, two parallel hyperplanes be selected to separate the two classes of data. The region bounded by these two hyperplanes is called the 'margin' and the points of x from both groups that are closest to the parallel hyperplanes are called 'support vector'. Here, x represents the feature vectors of the training data, which are essentially the attributes or characteristics of each data point used for classification. The maximum-margin hyperplane refers to the hyperplane situated exactly in the middle of these two, which is calculated from the distance of the support vector from each group perpendicular to the maximum-margin hyperplane with the assumption that the maximum-margin hyperplane should be as large as possible.

Predictions with the hyperplane utilise the following hypothesis equation

$$h(x) = \begin{cases} 1, \text{ if } \vec{w} \cdot x + b \geq 0 \\ -1, \text{ if } \vec{w} \cdot x + b < 0 \end{cases}$$

a set of points, represented as x_i (where i denotes the index of the data point and can range from 1 to n, with n being the total number of data points), is classified as 1, if the point is on or above the hyperplane or -1 if the point is on or below the hyperplane, which could be written as a new variable y_i; the variable $\hat{y}_i$ can be written as

$$\hat{y}_i = \begin{cases} 1, \text{ if } x_i \text{ is } on \text{ or } above \text{ the hyperplane} \\ -1, \text{ if } x_i \text{ is } on \text{ or } below \text{ the hyperplane} \end{cases}$$

Combining the equation above, then it could be rewritten as:

$$y_i(\vec{w} \cdot x_i + b) \geq 1 \qquad \text{for} \qquad 1 \leq i \leq n$$

$$\text{and } y_i(\vec{w} \cdot x_i + b) - 1 = 0 \quad \text{for } x_i \text{ as a support vector}$$

The principle above is referred to as a 'hard margin', which is used in cases where the data are fully separable from each other. In most circumstances, the data are noisy and the hyperplane cannot distinguish them unless the algorithm allows misclassification to occur, which

is defined as the use of a ‘soft margin’. To minimise errors, a loss function is added to the SVM algorithm, with hinge loss being the most common loss function.

Kernel Trick

In the case that the points of x from both outcomes overlap or have any outliers, the data will be non-separatable even when using soft margin to remove the noisy data problem. The kernel trick technique can be used for separating the data with SVM by transforming the data. It maps the data from the original space to a higher dimensional feature space, thereby making the data linearly separable in this new dimensional space.[85]

The kernel trick becomes very useful for multiple-dimensional data, where computation is more complicated. Instead of explicitly mapping data to a higher dimensional space, the kernel trick utilises kernel functions to compute inner products in the original feature space, effectively bypassing the need to work in the higher dimension. Taking two data points, x and y, in n dimensions as an example: without the kernel trick, the computations would require determining coordinates in an d^2 dimensional space. However, by employing a kernel function, denoted as $k(x, y)$, the same outcome is achieved within the d-dimensional space by computing the dot product of x-transpose and y. While various kernel functions exist, polynomial and polynomial and radial basis function (RBF) kernels are among the most prevalent. In this research, the RBF kernel was chosen for the SVM.

Data Processing and Model Development Tools

The extract, transform, and load (ETL) process was executed for MIMIC III data using SQL code on SQL Postgres Admin version 5.2. Model development was conducted using Python version 3.9.16, utilising the scikit-learn library for the LR, RF, and SVM models. All project code was uploaded to the author’s repository GitHub. See details in Appendix.

Model Performance Evaluation

Confusion Matrix

When evaluating the performance of a classification model, four fundamental metrics derived from the confusion matrix are often referenced:

		Actual Value	
Pre	dict	Positive	Negative

	Positive	True Positive (TP)	False Positive (FP)
	Negative	False Negative (FN)	True Negative (TN)

Table 4 Confusion matrix for classification model performance for binary outcomes.

- True Positives (TP): These are the cases in which the model correctly predicted the positive class. In other words, the actual class was positive, and the model predicted it as positive.
- True Negatives (TN): These are the cases in which the model correctly predicted the negative class. That is, the actual class was negative, and the model predicted it as negative.
- False Positives (FP), also known as Type I errors, are the cases in which the model incorrectly predicted the positive class. This means the actual class was negative, but the model predicted it as positive.
- False Negatives (FN), also known as Type II errors, are cases where the model incorrectly predicted the negative class. The actual class was positive, but the model predicted it as negative.

Evaluation Metrics

This study utilised accuracy and AUC as primary metrics to compare model performance.

Accuracy

The accuracy assesses overall performance by calculating the proportion of correct predictions relative to the total data. In other words, it is the ratio of TP and TN to the total data. However, this measure may be misleading and biased in imbalanced datasets. For instance, if a significantly smaller proportion of patients have an extended ICU LOS compared to those discharged earlier, relying solely on accuracy might not provide a comprehensive understanding of the model's performance. The accuracy formula has been shown as follows

$$Accuracy = \frac{TruePositive + TrueNegative}{TruePositive + FalsePositive + TrueNegative + FalseNegative}$$

Receiver Operating Characteristic (ROC)

The receiver operating characteristic (ROC) curve graphically depicts the binary classification model's performance by showing the balance between the true positive rate (TPR) or sensitivity and false positive rate (FPR) or 1-specificity at different thresholds. TPR refers to

the proportion of actual positives accurately identified, whereas FPR is the proportion of true negatives incorrectly labelled as positive.

Area Under the Curve (AUC)

The AUC, also known as the area under the receiver operating characteristic curve (AUROC), measures a model's ability to differentiate between positive and negative classes. It is calculated by determining the area under the ROC curve, for various decision thresholds. These thresholds define the points where probabilities transition between positive and negative class predictions. AUC values range from 0 to 1, where a value of 0 indicates a completely inaccurate test and a value of 1 reflects a perfectly accurate test. Generally, an AUC of 0.5 denotes to the absence of classification ability or mere random guessing, 0.7 – 0.8 is considered acceptable, a score from 0.8 to 0.9 is deemed excellent, and anything over 0.9 is regarded as exceptional classification.[86] However, the acceptability of AUC values can vary based on the specific classification task and context.

The AUC equation can be expressed as follows:

$$AUC = \int_0^1 \big(TPR(FPR)\big) dFPR$$

TPR(FPR) represents the TPR at a given FPR value.

AUC is a preferred metric because it is insensitive to class imbalance and can provide a global summary of the model's performance across all possible thresholds. It encapsulates the trade-offs between sensitivity and specificity, making it a more informative metric than accuracy alone, especially in contexts where both types of errors are equally important.

Five-Fold Cross-Validation

Five-fold cross-validation is a commonly used technique in machine learning and model selection for estimating the performance of an estimator or for performing model selection by dividing the dataset into five subsets or ‘folds’.[87] Five-fold cross-validation involves the following steps.

First, the dataset is randomly divided into five equal-sized subsets. Four of these subsets serve as the training set, and the remaining subset, known as the validation set, is used to evaluate the model's performance. This process of training and validation is repeated five times,

with a different subset serving as the validation set each time. The performance metrics, namely, accuracy and AUC, are determined by averaging the results from the five iterations.

Five-fold cross-validation could provide a more reliable estimate of the model's performance than a single train-test split. It helps to reduce the risk of overfitting or underfitting by mitigating the impact of specific data points in the training and validation sets. While five-fold cross-validation does enhance our evaluation by repeating the process with varying subsets, it provides only a partial assessment of the model's generalisation ability.[87] True generalisation is ultimately demonstrated when the model is applied to an entirely new dataset.

The choice of five-fold cross-validation over other variations, such as ten-fold or leave-one-out depends on the specific features of the problem, including dataset size, model complexity, and the presence of outliers or class imbalance. In this case, five-fold was selected to minimise overfitting with the MIMIC III dataset.

Feature Importance

Feature importance refers to the technique used to determine the relative significance or influence of each feature in a predictive model. This process helps in comprehending which attributes have the most influence on the outcome depend on the model performance. By identifying the most important features, researchers can gain insights into the fundamental factors affecting the model's predictions.[88]

Different algorithms calculate feature importance using different methods. Therefore, the calculated feature importance value is unsuitable for cross-algorithm comparison. Instead, it is only useful for evaluating the relevance of attributes within the specific algorithm itself.

Logistic Regression (LR)

A common approach when using an LR model is to use the coefficients of the model, which signify the alteration in the log-odds of the outcome for each one-unit change in the corresponding input feature. Larger coefficient values denote a more pronounced influence on the prediction.[89]

Random Forest (RF)

The feature importance in RF is from its base classifier, the DT. For each tree in the forest, feature importance is determined by how often a feature is used to split the data and by the average reduction in the Gini index caused by that feature. Essentially, if a feature consistently

results in nodes with high purity (larger drops in the Gini index), it is deemed important. In RF, the overall feature importance for each feature is averaged across all trees in the forest.[90]

Support Vector Machine (SVM)

The weights drawn from the SVM coefficients represent the vector coordinates perpendicular to the hyperplane. The coordinates of the vector define the orientation and position of the hyperplane that SVM utilises to separate data points of different classes. These coordinates are perpendicular to the hyperplane, and their direction indicates the predicted class. The absolute value of the coefficients is related to the importance of each feature when separating the data.[84] However, these coefficients apply only to SVM with a linear kernel. For other kernels, such as polynomial or radial basis function kernels, the data are transformed by the kernel method to a different space that is unrelated to the input space. Therefore, the coefficients cannot be applied, as the data are transformed by the kernel method to a space unrelated to the input space.[84] For non-linear SVMs, alternative methods, such as permutation feature importance, can be employed to assess feature importance.

Results

Exploratory Data Analysis

The MIMIC-III dataset contains a total of 58,976 unique hospital admission cases, which involve 46,520 unique patients. After applying the ICD-9 primary diagnosis for stroke to the dataset, 6,220 hospital admission cases involving 4,873 unique patients were identified.

Following the exclusion criteria, 670 elective cases, 16 cases where patients were below 20 years old, and 497 cases with TIA and precerebral or vertebral vascular pathology were removed, which resulted in 5,576 remaining hospital admission cases involving 4,237 unique patients. Further, 2,277 cases associated with patient deaths during admission were removed. Additionally, 13 variables (body temperature, respiratory rate, serum creatinine, triglycerides, cholesterol, LDL, total cholesterol, total bilirubin, albumin, hemoglobin A1c, $PaCO_2$, PaO_2, pH, and lactate), as well as 1,428 case records, were eliminated due to missing data exceeding 20% of the total data. Table 3 displays the number of missing values per remaining feature.

Thereafter, only the first-time ICU admissions within the same admission was selected, resulting in 555 cases of unique ICU admission and unique patients. Fig 2 shows the process of data extraction.

The variable 'Time from ER to ICU' was also removed. Initially, it was to be calculated by subtracting the hospital admission time from the ICU admission time. However, shifting admission dates in the MIMIC III dataset[70], a measure taken to de-identify patients over 80 years old, compromised the reliability of this variable. The shift resulted in an abnormal range – between -20,000 and 8,000,000 waiting minutes for ICU admission – which necessitated its exclusion. After these adjustments, 555 case records and 34 variables remained for the training and testing dataset.

Variables	Missing Values (%)
Age Group	0
Gender	0
Ethnicity Grouped	0
Insurance	0
Admission Type	0
History of Previous Stroke	0
Ventilator Status	0

Variables	Missing Values (%)
ICU LOS	0
Hematocrit	0
Hemoglobin	0
Lymphocytes	13.28
Neutrophils	13.28
Platelet Count	0
White Blood Cells	0
INR(PT)	0.49
PT	0.46
PTT	0.56
ALT	17.69
ALP	19.31
AST	17.62
Bicarbonate	0
Chloride	0
Glucose	0
Potassium	0
Sodium	0
Urea Nitrogen	0
Admission Weight	15.71
GCS - Eye Opening	15.78
GCS - Motor Response	15.85
GCS - Verbal Response	15.82
Heart Rate	15.64
Diastolic Blood Pressure	16.28
Systolic Blood Pressure	16.28
O_2 Saturation Pulse Oximetry	15.64

Table 5 Information on missing values per feature.

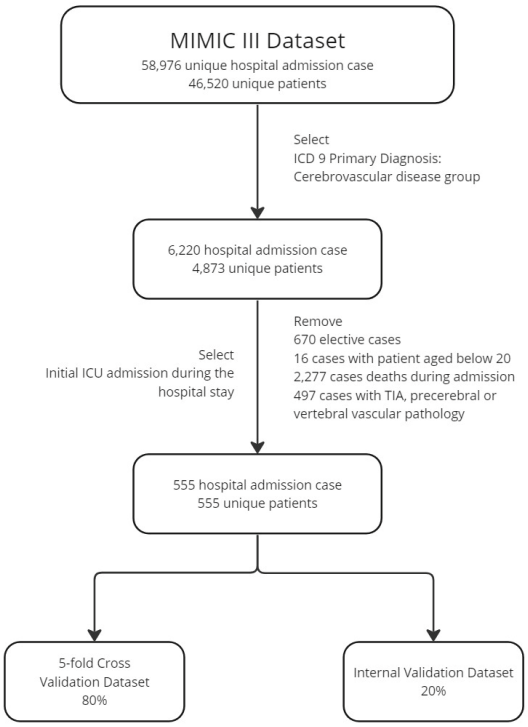

Figure 2 The process of MIMIC III data extraction.

Descriptive Analysis

Assessing Normality

According to the Shapiro–Wilk statistic, almost all numeric variables exhibited a p-value of less than 0.05, signifying statistical significance. This suggests that these variables might not follow a normal distribution. This information is crucial as it guides the approach to summarising data and addressing missing values. Refer to Appendix Table A for detailed results.

Summary of ICU LOS

Fig 3 presents the ICU LOS histogram for this study. The data exhibits a right-skewed distribution. The median ICU LOS is 4 days, with an interquartile range (IQR) spanning from 2 to 7 days. Notably, ICU LOS does not exceed two weeks for 90% of the observations.

Univariable Analysis

The univariable analysis was performed after a single imputation process. The study consists of 555 patients, with a slightly higher proportion of males (54.6%) than females (45.4%). The majority of patients fell within the age groups of over 80 years (36.6%) and 70–79 years (28.5%). Most patients were of White ethnicity (67.2%) and were covered by Medicare insurance (70.6%).

The predominant admission type was emergency (98.6%), and the majority (83.6%) required respiratory support during their stay. The most common ICD-9 Code Diagnosis was 434 for 'occlusion of cerebral arteries' at 35.5%. Further information of univariable analysis can be found in Table 4.

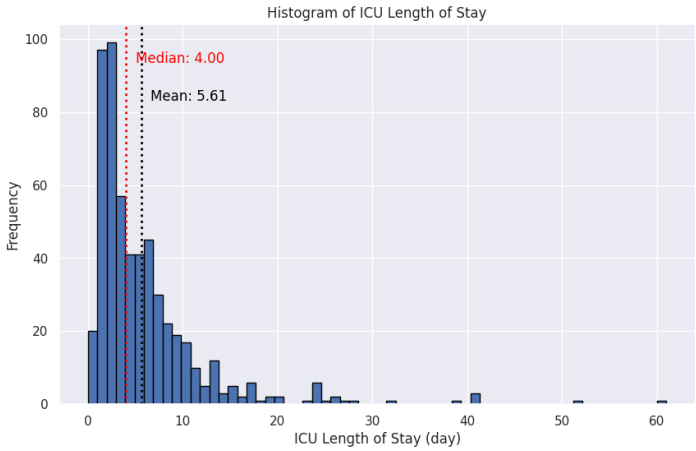

Figure 3 Histogram of ICU LOS with the x axis representing the number of ICU LOS and the y axis representing frequency.

Number of Patients			555
Demographic	Gender (n [%])		
		Male	303 [54.6]
		Female	252 [45.4]
	Age Group: Year (n [%])		
		20-29	4 [0.7]
		30-39	8 [1.4]
		40-49	24 [4.3]
		50-59	67 [12.1]
		60-69	91 [16.4]
		70-79	158 [28.5]
		over 80	203 [36.6]
	Ethnicity Grouped (n [%])		

		Asian	19 [3.4]
		Black	42 [7.6]
		Hispanic	20 [3.6]
		Other	18 [3.2]
		Unknown	83 [15.0]
		White	373 [67.2]
	Insurance (n [%])		
		Government	4 [0.7]
		Medicaid	41 [7.4]
		Medicare	392 [70.6]
		Private	112 [20.2]
		Self-Pay	6 [1.1]
Admission Information	Admission Type (n [%])		
		Emergency	547 [98.6]
		Urgency	8 [1.4]
	History of Previous Stroke (n [%])		
		No	545 [98.2]
		Yes	10 [1.8]
	ICU Length of Stay: Day (median [IQR])		4.00 [2.00, 7.00]
	ICU Length of Stay: Category (n [%])		
		≤ 24 hrs.	20 [3.60]
		1-6 days	380 [68.47]
		7-13 days	115 [20.72]
		≥ 14 days	40 [7.21]
	ICD-9 Code Diagnosis Group (n [%])		
		430 Subarachnoid haemorrhage	60 [10.8]
		431 Intracerebral haemorrhage	152 [27.4]
		432 Other and unspecified intracerebral haemorrhage	47 [8.5]
		433 Occlusion and stenosis of precerebral arteries	52 [9.4]
		434 Occlusion of cerebral arteries	197 [35.5]
		436 Acute but ill-defined cerebrovascular disease	2 [0.4]
		437 Other and ill-defined cerebrovascular disease	45 [8.1]
	Respiratory Support Status (n [%])		
		No	91 [16.4]
		Yes	464 [83.6]
Laboratory Test: Complete Blood Count	Haematocrit: % (median [IQR])		32.80 [28.00, 37.30]
	Haemoglobin: g/dL (median [IQR])		11.10 [9.30, 12.70]
	White Blood Cells: x 10^3/μL (median [IQR])		11.80 [9.20, 15.90]
	Neutrophils: % (median [IQR])		84.6 [76.70, 88.90]

	Lymphocytes: % (median [IQR])		9.0 [5.90, 15.00]
	Platelet Count: x 10^3/μL (median [IQR])		219.00 [154.00, 289.50]
Laboratory Test: Coagulation	INR(PT) (median [IQR])		1.20 [1.10, 1.50]
	PT: Second (median [IQR])		13.70 [12.70, 15.90]
	PTT: Second (median [IQR])		28.30 [24.20, 36.00]
Laboratory Test: Clinical Chemistry	ALT: IU/L (median [IQR])		29.00 [17.75, 49.00]
	ALP: IU/L (median [IQR])		79.00 [61.00, 116.50]
	AST: IU/L (median [IQR])		39.0 [26.0, 72.0]
	Bicarbonate: mEq/L (median [IQR])		24.00 [21.00, 27.00]
	Chloride: mEq/L (median [IQR])		79.00 [61.00, 116.50]
	Glucose: mg/dL (median [IQR])		145.00 [118.00, 180.00]
	Potassium: mEq/L (median [IQR])		3.90 [3.60, 4.35]
	Sodium: mEq/L (median [IQR])		140.00 [137.00, 145.00]
	Urea Nitrogen: mEq/L (median [IQR])		24.00 [16.00, 35.00]
Physical Examination at Admission	Admission Weight: kg (mean (SD))		74.70 [61.00, 89.00]
	Glasgow Coma Scale		
		GCS - Eye Opening (median [IQR])	1.00 [1.00, 3.00]
		GCS - Verbal Response (median [IQR])	1.00 [1.00, 1.00]
		GCS - Motor Response (median [IQR])	4.00 [1.00, 5.00]
	Heart Rate: bpm (median [IQR])		85.00 [72.00, 98.25]
	Diastolic Blood Pressure: mmHg (median [IQR])		61.00 [51.00, 73.00]
	Systolic Blood Pressure: mmHg (median [IQR])		124.48 [108.5, 139.0]
	Oxygen Saturation Pulse Oximetry: % (median [IQR])		98.00 [96.39, 100.0]

Table 6 The univariate analysis information for variables utilised in model prediction and ICU LOS

ALP = Alkaline Phosphatase, ALT = Alanine Aminotransferase, AST = Aspartate Aminotransferase, bpm = beat per minute, C° = Celsius ,cm = centimetres, dL = Decilitres, g= Gram, GCS = Glasgow Coma Scale, INR = International normalised ratio, IQR = Interquartile Range, IU = International unit, kg = Kilogram, L = Liters, LOS = Length of stay, mEq = milliequivalents, mg = Milligram, ml = Millimetre, mmHg = Millimetre of mercury, mmole = Millimole, PT = Prothrombin time, PTT = Partial thromboplastin time, SD = Standard Deviation, μL = Microliters

Correlation

Spearman's correlation among the quantitative variables, which include both numerical and ordinal variables, is illustrated in Fig 3a. The most negative correlation was -0.7971, observed between the neutrophils and lymphocytes variables, and the maximum value was 0.7680, observed between aspartate aminotransferase and alanine aminotransferase. Although no values exceeded 0.8, values close to this threshold, such as 0.7971, suggest potential multicollinearity, which may warrant further examination.

The Cramer V statistic is shown in Fig 4B., with a minimum value of 0 and a maximum value of 0.37. The maximum value represents the association between the age group and

insurance. See the supplementary section for additional details. While no associations exceed a commonly considered threshold for strong multicollinearity, potential relationships between any of the numerical and ordinal variables or categorical variables may warrant further examination.

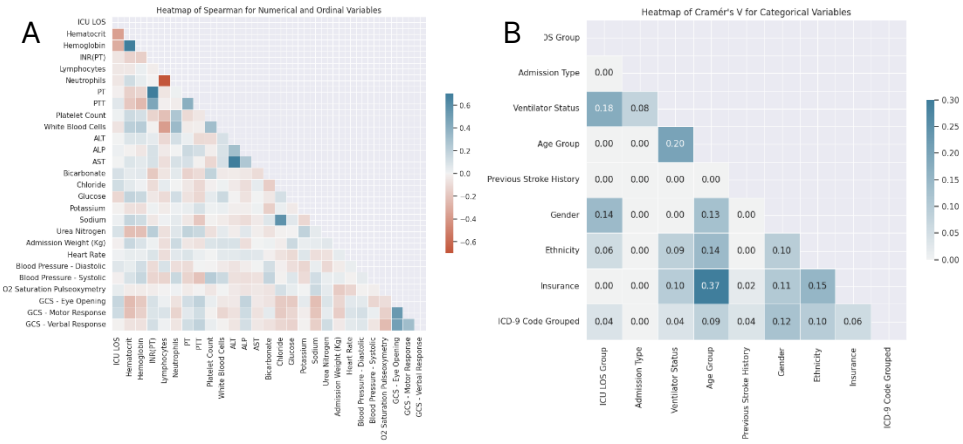

Figure 4 Correlation statistic 4A. Spearman statistics for numerical and ordinal data, 4B. Cramér's V statistic for categorical variables.

Model Prediction Performance

The performance metrics for ICU LOS classification models are presented in Table 4. During the 5-fold cross-validation, the RF model exhibited the highest accuracy with values of 0.82 with SE of 0.025 for accuracy, as well as 0.82 with SE of 0.037 for AUC. For internal validation, the RF model again achieved superior results with an accuracy of 0.81 and an AUC of 0.85. The LR and SVM models demonstrated similar performances, with slight variations in their accuracy and AUC values. When compared to the results from Kurtz, P., et al., our models generally showed comparable or better AUC values. The results of LR is reported with threshold default of 0.5 as it provided better performance in overall.

	5-fold Cross Validation		Internal Validation		Kurtz, P., et al.[52]
Model	ACC (Mean, SE)	AUC (Mean, SE)	ACC	AUC	AUC
LR	0.75, 0.022	0.72, 0.065	0.76	0.74	0.71
RF	0.82, 0.025	0.82, 0.037	0.81	0.85	0.73
SVM	0.75, 0.025	0.70, 0.072	0.74	0.75	0.68

Table 7 Comparison of accuracy and AUC of ICU LOS classification models with previous work by Kurtz, P., et al. (2022).

ACC = Accuracy, AUC = Area Under the Curve, LR = Logistic Regression, RF = Random Forest, SE = Standard Error, SVM = Support Vector Machine

Feature Importance

In the evaluation of feature importance for predicting ICU LOS across three models, several patterns emerge. RF model, which delivered the best performance, emphasizes haematocrit, platelet count, and haemoglobin as the top three influential features. In the SVM model, INR(PT), Ethnicity: other, and Insurance: Government dominate the feature significance. In contrast, LR model points to INR(PT), Ethnicity: other, and Ventilator Status: 0 (No) as top three importance. A comparison of the top 15 features across all models reveals that haematocrit, INR(PT), and neutrophils are recurrently influential. However, the models demonstrate varied emphasis on certain features, suggesting that the choice of model can affect the perceived importance of predictors in ICU LOS outcomes. Fig 4 illustrates the top 15 features for each model. Additional details are available in the Supplementary section.

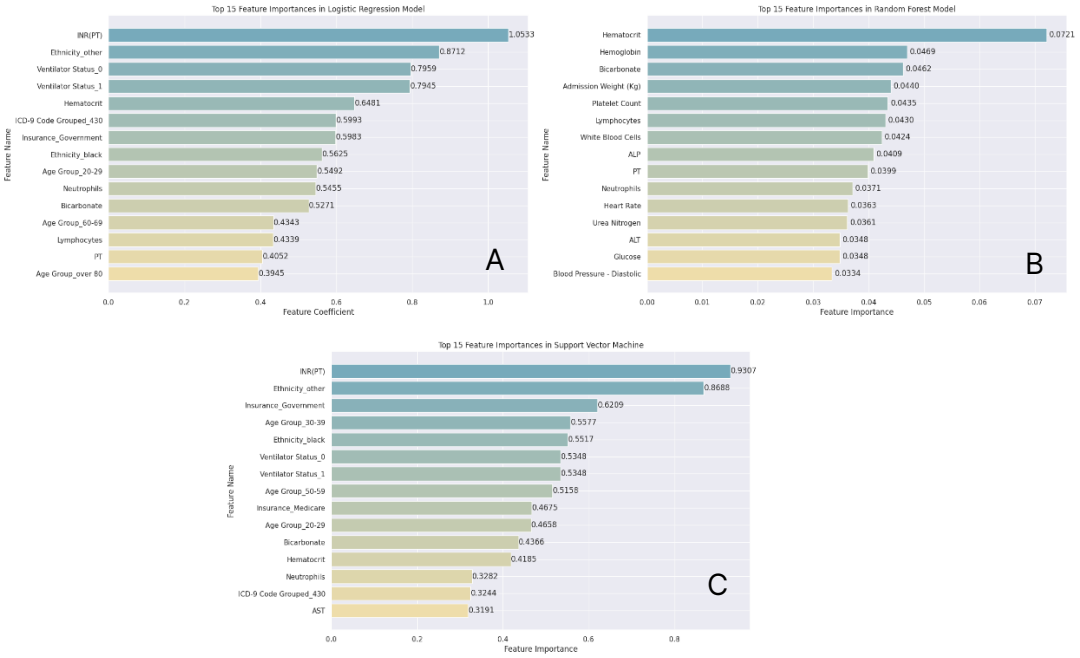

Figure 5 Top 15 feature importance for 4A. Logistic regression model, 4B. Random Forest model and 4C. Support vector machine model.

Discussion

Main Findings

This study aimed to predict ICU LOS for stroke patients, a critical factor in healthcare resource allocation and patient outcomes. Using the MIMIC-III dataset, the study developed a machine learning-based model to predict ICU LOS for individuals diagnosed with stroke. Among the three classification algorithms evaluated: LR, RF, and SVM. The RF model proved most effective. It achieved an accuracy of 0.82 in 5-fold cross-validation and 0.81 during internal validation. These results highlight the model's potential as a valuable planning tool for decision-making in ICU management, especially in stroke patient care.

Considerations and Implications of Excluding Mortality Data in ICU LOS Analyses

ICU LOS primarily focuses on the duration a patient stays in intensive care before being either discharged from the ICU or transferred to another level of care. Including patients who died during ICU admission can distort this measure, as their length of stay might be influenced by the severity of their condition rather than the typical course of treatment or recovery for the condition being studied. In this study, the records of patients who died during hospital admission were excluded. This decision aligns with many previous studies where ICU LOS analyses often did not explicitly account for mortality. As noted by the systematic review study by Harhay, et al. (2019), a significant proportion of critical care trials have either ignored mortality entirely or stratified analyses by survival status when comparing ICU LOS. Such approaches can lead to ambiguous or misleading conclusions [91], for example, if a data that usually represent in critically severe patient (e.g., unstable vital sign or treatment for life resuscitation), it may appear as if the data might predict in a shorter ICU stay, even though the outcome is unfavourable. For final dataset, there were 555 patients who were alive during admission, providing a specific representation of ICU LOS for surviving patients. On the other hand, the 2,277 cases of patients who died during admission could skew the ICU LOS metrics. If the data were incorporated into the analysis, the average ICU LOS might be artificially shortened by those who died shortly after admission or extended by those who remained in the ICU for longer durations before passing away. Such a representation could lead to incorrect inferences about the typical length of ICU stays for treatment and recovery.

Relationships Among Variables

This project had multiple objectives. Initially, it aimed to explore associations between features and their relationship with ICU length of stay. The Spearman's and Cramer V statistics showed that, although no values exceeded the multicollinearity threshold, some were close. Although this study did not take specific action on the near-threshold multicollinearity, future research should consider techniques such as ridge regression or variable elimination to better understand the influence of predictors.

The study also had the issue of missing data, which led to the exclusion of a large amount of data. The loss of data points could potentially impact associations between features and their relationship with ICU length of stay.

Model Performance and Comparative Analysis

Another aim of the project is to leverage machine learning models to estimate ICU length of stay (LOS) for stroke patients using the MIMIC-III Critical Care Database. Of the models assessed, including LR, RF, and SVM, the RF model consistently outperformed the others in cross-validation and internal validation. When compared with the findings of Kurtz, P., et al. (2022), models in this study show better result in accuracy and similar or improved AUC values. This emphasizes the potential of machine learning in ICU LOS prediction and highlights the significance of choosing a suitable model tailored to specific datasets and goals.

Upon closer examination, the enhanced performance of the models in this study compared to the Kurtz study can be attributed to a more comprehensive feature set. This study incorporated laboratory test data, including variables such as haematocrit, INR(PT), and neutrophils, which proved as important factors in all models. This additional data likely improved the model's predictive accuracy. In contrast, the Kurtz study focused solely on demographic information, ICU admission diagnoses, comorbidities, and medical assessment scores, omitting vital signs and laboratory test data. The inclusion of these omitted variables in this study likely accounts for its superior performance.

Rationale for Using the Default Threshold of 0.5 in Logistic Regression

In logistic regression, the default threshold value of 0.5 is commonly used for classifying an observation into one of the two classes. This assumes that both classes are equally important, and the costs of false positives and false negatives are roughly the same. However,

in some cases, adjusting the threshold could potentially improve model performance as evaluated by specific metrics such as accuracy, precision, recall, or the area under the ROC curve (AUC). In this instance, an "optimal" threshold of 0.784725 was identified, but the results were mixed. Although the AUC from 5-fold cross-validation was slightly higher with the optimal threshold compared to the default (0.744 vs. 0.72), the accuracy on internal validation was notably lower (0.694 vs. 0.76). Additionally, the overall performance did not surpass that of a Random Forest model. Given these considerations, and because the performance metrics using the default threshold were more consistent across both training and validation sets, the author decided to report performance using the default threshold of 0.5. This choice also makes the results more interpretable and easily comparable with other models and studies that use the conventional 0.5 threshold. See details in Appendix.

Limitations

This study has several limitations. First, the detailed data extraction from the MIMIC-III Critical Care Database. The finalized dataset of 555 ICU admissions, while focusing on stroke patients, may have missed critical insights due to the strict exclusion criteria and omission of certain variables. Notably, the removal of the 'Time from ER to ICU' variable, resulting from data shifting measures, restricted our understanding of ICU patient flow dynamics. Additionally, the proximity of certain Spearman's correlation values to multicollinearity thresholds indicates intricate predictor relationships. Given that the findings predominantly rely on the RF model's performance within a single database, their applicability might be constrained in different healthcare contexts or patient groups. Another limitation is the data imbalance in the dataset. Even though efforts were made to maintain a consistent ratio during data splitting, the inherent imbalance between the ICU LOS predictions of less than 7 days and more than 7 days may impact the model's ability to generalize its predictions effectively across diverse scenarios. Imbalanced datasets can lead to models that are biased towards the majority class, which can affect their overall performance and reliability.

Recommendations for Future Research

Based on the project result and its limitations. Several suggestions could be implemented for the future work related to ICU LOS prediction in stroke patient.

1. Database Expansion: To ensure comprehensive analysis and enhance result generalizability, it is recommended to utilize data from various sources, not limited to the MIMIC III Critical Care Database.

2. Inclusion of Previously Excluded Variables: The exclusion of the 'Time from ER to ICU' variable, due to de-identification concerns, might have limited certain insights. Future research could explore alternative databases or methods to incorporate this variable while ensuring patient confidentiality.

3. Refined Data Imputation: Instead of solely using a single imputation method, adopting multiple imputation techniques, such as multiple imputation by chained equations (MICE) could offer a more robust approach to handling skewed or missing data. This might improve the model's performance.

4. Addressing Multicollinearity: The dataset exhibited signs of potential multicollinearity. Future endeavours should closely analyse these relationships like variance inflation factor (VIF) analysis or principal component analysis (PCA) to identify and mitigate multicollinearity, particularly when incorporating new datasets.

5. Exploration of Alternative Algorithms: While the RF model was effective in this study, examining other machine learning or ensemble techniques might identify models with enhanced prediction accuracy.

6. Expanding Demographic Representation: Given the noted dominance of White ethnicity in the dataset, future studies should aim for a diverse patient representation, covering varied ethnicities, ages, and medical histories.

7. Intensive Analysis of Key Features: Features such as haematocrit, INR(PT), and neutrophils were consistently influential across models. Investigating the clinical significance of these features in the context of ICU LOS is warranted.

Adhering to these recommendations can improve the effectiveness of future research, refining the predictive power of machine learning models for ICU LOS and the improved predictive performance could serve as a valuable tool for data-driven decision-making in resource allocation and treatment planning.

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Appendix

All project code was uploaded to the author's repository GitHub: https://github.com/kanyakornvee/LSHTM_MIMIC.

Variable	p-value
ICU LOS: Day	<0.0001
Haematocrit	0.0004
Haemoglobin	0.0004
INR(PT)	<0.0001
Lymphocytes	<0.0001
Neutrophils	<0.0001
PT	<0.0001
PTT	<0.0001
Platelet Count	<0.0001
White Blood Cells	<0.0001
ALT	<0.0001
ALP	<0.0001
AST	<0.0001
Bicarbonate	<0.0001
Chloride	<0.0001
Glucose	<0.0001
Potassium	<0.0001
Sodium	<0.0001
Urea Nitrogen	<0.0001
Admission Weight (Kg)	<0.0001
GCS - Eye Opening	<0.0001
GCS - Motor Response	<0.0001
GCS - Verbal Response	<0.0001
Heart Rate	<0.0001
Non-Invasive Blood Pressure diastolic	<0.0001
Non-Invasive Blood Pressure systolic	0.0009
Oxygen Saturation	<0.0001

Supplementary Table A. P-value from the Shapiro-Wilk correlation for numeric variables.

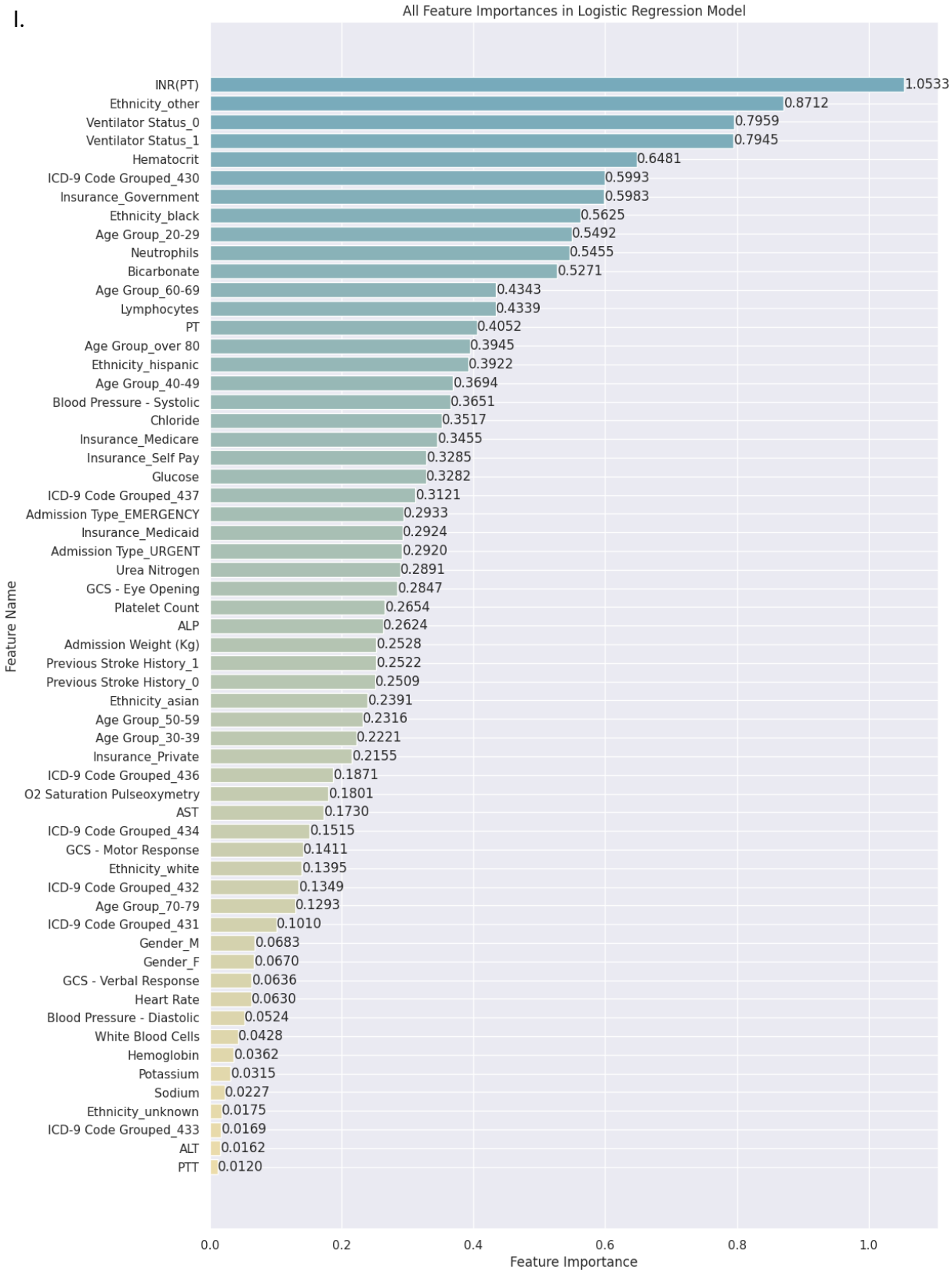
Age Group	Government	Medicaid	Medicare	Private	Self-Pay
20-29	1	2	0	1	0
30-39	1	2	0	5	0
40-49	0	7	4	12	1
50-59	2	18	8	36	3
60-69	0	9	53	27	2
70-79	0	2	133	23	0
over 80	0	1	194	8	0

Supplementary Table B. Patient distribution by age group and insurance types.

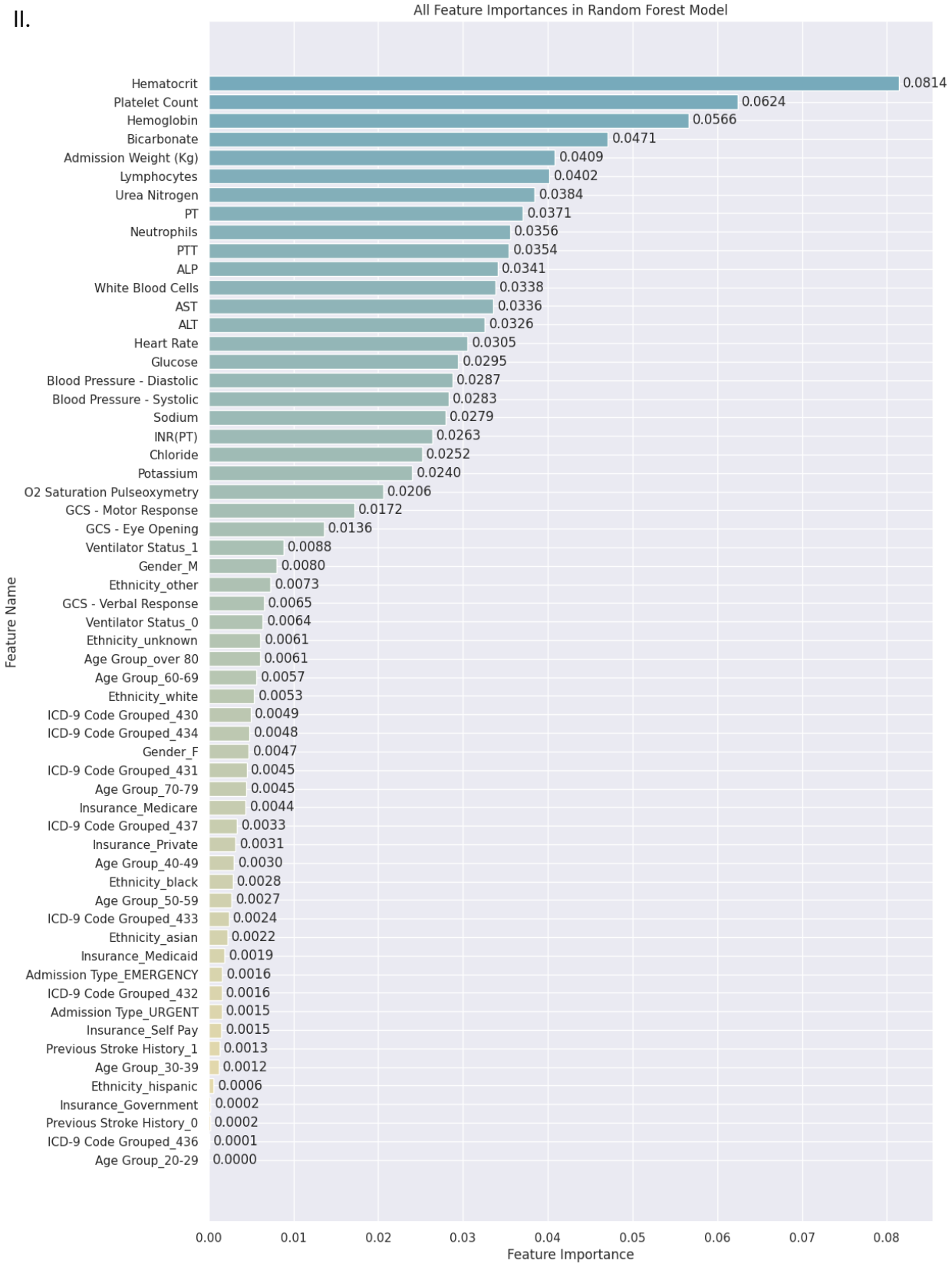
	5-fold Cross Validation		Internal Validation	
Model	ACC (Mean, SE)	AUC (Mean, SE)	ACC	AUC
LR Default threshold (0.5)	0.75, 0.022	0.72, 0.065	0.76	0.74
LR Optimal threshold (0.78)	0.75, 0.013	0.74, 0.025	0.69	0.74

Supplementary Table C. Accuracy and AUC of ICU LOS classification models with optimal threshold for logistic regression model.

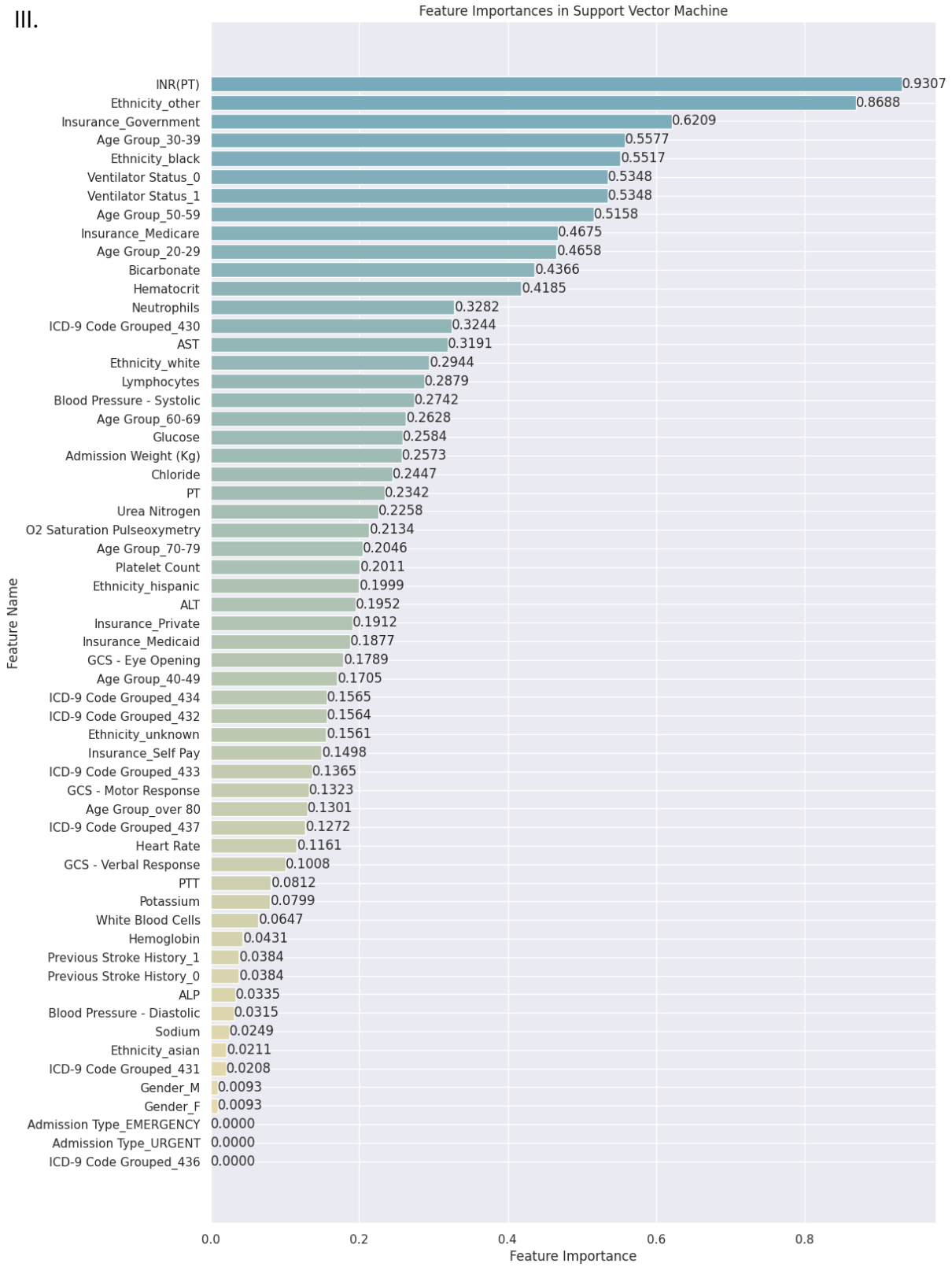
I.


All Feature Importances in Logistic Regression Model
INR(PT) 1.0533
Ethnicity_other 0.8712
Ventilator Status_0 0.7959
Ventilator Status_1 0.7945
Hematocrit 0.6481
ICD-9 Code Grouped_430 0.5993
Insurance_Government 0.5983
Ethnicity_black 0.5625
Age Group_20-29 0.5492
Neutrophils 0.5455
Bicarbonate 0.5271
Age Group_60-69 0.4343
Lymphocytes 0.4339
PT 0.4052
Age Group_over 80 0.3945
Ethnicity_hispanic 0.3922
Age Group_40-49 0.3694
Blood Pressure - Systolic 0.3651
Chloride 0.3517
Insurance_Medicare 0.3455
Insurance_Self Pay 0.3285
Glucose 0.3282
ICD-9 Code Grouped_437 0.3121
Admission Type_EMERGENCY 0.2933
Insurance_Medicaid 0.2924
Admission Type_URGENT 0.2920
Urea Nitrogen 0.2891
GCS - Eye Opening 0.2847
Platelet Count 0.2654
ALP 0.2624
Admission Weight (Kg) 0.2528
Previous Stroke History_1 0.2522
Previous Stroke History_0 0.2509
Ethnicity_asian 0.2391
Age Group_50-59 0.2316
Age Group_30-39 0.2221
Insurance_Private 0.2155
ICD-9 Code Grouped_436 0.1871
O2 Saturation Pulseoxymetry 0.1801
AST 0.1730
ICD-9 Code Grouped_434 0.1515
GCS - Motor Response 0.1411
Ethnicity_white 0.1395
ICD-9 Code Grouped_432 0.1349
Age Group_70-79 0.1293
ICD-9 Code Grouped_431 0.1010
Gender_M 0.0683
Gender_F 0.0670
GCS - Verbal Response 0.0636
Heart Rate 0.0630
Blood Pressure - Diastolic 0.0524
White Blood Cells 0.0428
Hemoglobin 0.0362
Potassium 0.0315
Sodium 0.0227
Ethnicity_unknown 0.0175
ICD-9 Code Grouped_433 0.0169
ALT 0.0162
PTT 0.0120
Feature Name
0.0
0.2
0.4
0.6
0.8
1.0
Feature Importance

II.


All Feature Importances in Random Forest Model
Feature Name
Feature Importance
Hematocrit 0.0814
Platelet Count 0.0624
Hemoglobin 0.0566
Bicarbonate 0.0471
Admission Weight (Kg) 0.0409
Lymphocytes 0.0402
Urea Nitrogen 0.0384
PT 0.0371
Neutrophils 0.0356
PTT 0.0354
ALP 0.0341
White Blood Cells 0.0338
AST 0.0336
ALT 0.0326
Heart Rate 0.0305
Glucose 0.0295
Blood Pressure - Diastolic 0.0287
Blood Pressure - Systolic 0.0283
Sodium 0.0279
INR(PT) 0.0263
Chloride 0.0252
Potassium 0.0240
O2 Saturation Pulseoxymetry 0.0206
GCS - Motor Response 0.0172
GCS - Eye Opening 0.0136
Ventilator Status_1 0.0088
Gender_M 0.0080
Ethnicity_other 0.0073
GCS - Verbal Response 0.0065
Ventilator Status_0 0.0064
Ethnicity_unknown 0.0061
Age Group_over 80 0.0061
Age Group_60-69 0.0057
Ethnicity_white 0.0053
ICD-9 Code Grouped_430 0.0049
ICD-9 Code Grouped_434 0.0048
Gender_F 0.0047
ICD-9 Code Grouped_431 0.0045
Age Group_70-79 0.0045
Insurance_Medicare 0.0044
ICD-9 Code Grouped_437 0.0033
Insurance_Private 0.0031
Age Group_40-49 0.0030
Ethnicity_black 0.0028
Age Group_50-59 0.0027
ICD-9 Code Grouped_433 0.0024
Ethnicity_asian 0.0022
Insurance_Medicaid 0.0019
Admission Type_EMERGENCY 0.0016
ICD-9 Code Grouped_432 0.0016
Admission Type_URGENT 0.0015
Insurance_Self Pay 0.0015
Previous Stroke History_1 0.0013
Age Group_30-39 0.0012
Ethnicity_hispanic 0.0006
Insurance_Government 0.0002
Previous Stroke History_0 0.0002
ICD-9 Code Grouped_436 0.0001
Age Group_20-29 0.0000
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
0.08

III.



Supplementary Figure A. All feature importance values for AI. logistic regression model, AII. Random Forest model and AIII. support vector machine model